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Prepared By	iNgabo Platform Architecture Team, Rwanda Coding Academy
Technical Lead	Senior Solution Architect, iNgabo Platform
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EXECUTIVE SUMMARY



ES.1 Document Purpose

This Technical Architecture Document (TAD) describes the complete engineering design, system implementation, component architecture, data flows, AI subsystems, telecom integration, security architecture, and investigation workflows for iNgabo - Rwanda's National Telecom Fraud Intelligence, Citizen Protection, and Device Recovery Platform.

This document is the authoritative technical reference for Rwanda Investigation Bureau (RIB) engineering evaluation, RURA technical compliance review, NCSA security assessment, and for engineering teams implementing and extending the platform.

ES.2 Platform Overview

iNgabo is a purpose-built national digital infrastructure platform integrating six operational domains into a unified intelligence and investigation environment:

Table 1: iNgabo Operational Domain Summary

Domain	Capability	Primary Beneficiary
Phone Theft & Recovery	IMEI-based device tracking, SIM swap detection, network-assisted recovery, chain-of-custody management	RIB, Rwanda National Police, Citizens
Telecom Fraud Detection	Real-time SS7/DIAMETER anomaly detection, SIM cloning, number spoofing, account takeover	RURA, MNOs, Financial Institutions
SMS Intelligence	NLP-based smishing, phishing, URL threat analysis, bulk scam campaign detection	Citizens, MNOs, BNR
Voice Call Intelligence	Voice metadata analysis, vishing detection, caller reputation, social engineering indicators	Citizens, RIB, RNP
Police & Investigation Dashboards	Case management, fraud network visualization, location intelligence, AI recommendations	RIB, RNP, Cybercrime Unit
Citizen Security Platform	Fraud reporting, number verification, IMEI check, proactive alerts, case tracking	All Rwandan Citizens

ES.3 Architectural Approach

iNgabo is implemented as a cloud-native, event-driven microservices platform deployed on Kubernetes. The architecture applies Zero Trust security principles throughout, with no implicit trust between components, mandatory mutual TLS, and continuous authorization enforcement. The AI/ML subsystem combines supervised fraud classification, unsupervised behavioral anomaly detection, graph neural network analysis on Neo4j, and NLP-based content classification - all with mandatory Explainable AI (XAI) output and Human-in-the-Loop (HITL) controls for enforcement actions.

ES.4 CAMARA API Integration

The platform integrates GSMA CAMARA open network APIs to leverage mobile network capabilities including Device Location, SIM Swap Detection, Number Verification, Device Status, and KYC Match - providing telecom-signal-backed fraud intelligence that exceeds what application-layer analysis alone can deliver. All CAMARA integrations operate under formal data sharing agreements and regulatory authorization from RURA and applicable MNOs.

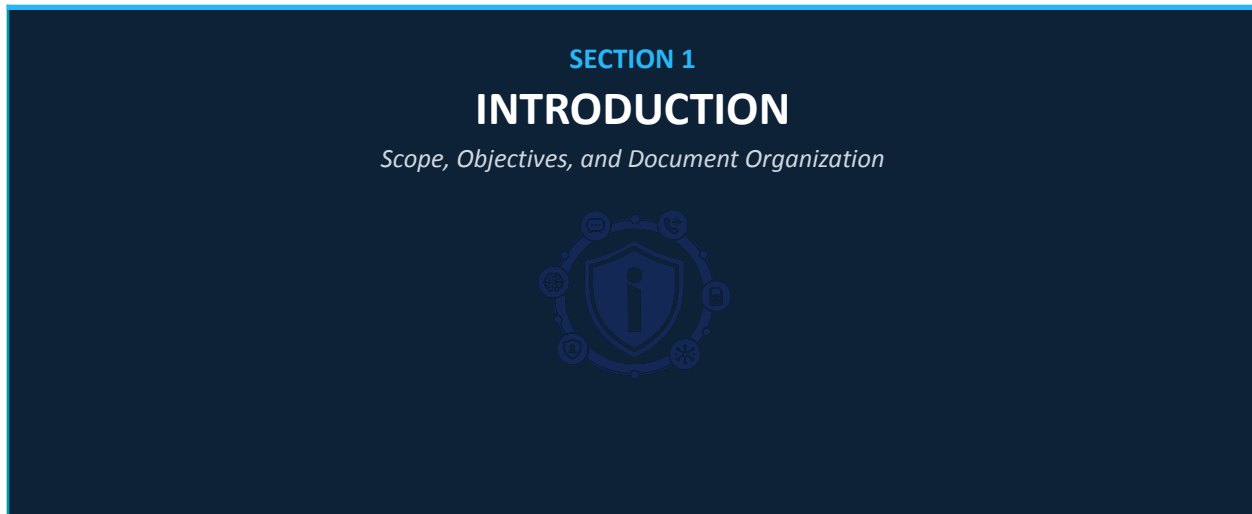
ES.5 Key Performance Targets

Table 2: iNgabo Platform Performance Targets

Metric	Target	Notes
System availability (fraud detection core)	99.99%	Active-active multi-zone deployment
Fraud alert latency (real-time path)	< 2 seconds	From signal ingestion to alert dispatch
Risk score API response (P95)	< 500 ms	Under nominal production load

Metric	Target	Notes
SIM swap intercept latency	< 1 second	Gateway intercept before HLR confirmation
SMS classification throughput	> 50,000 messages/sec	Horizontal scaling on Kafka/ML workers
Graph query response (fraud network)	< 3 seconds	Neo4j cluster with query caching
Citizen portal response	< 2 seconds	CDN-fronted, Redis-cached responses
Investigation dashboard load	< 4 seconds	Full case graph with 1,000-node networks

1. INTRODUCTION



1.1 Purpose

This document specifies the technical architecture of iNgabo v2.0. It provides sufficient engineering detail for:

- Technical evaluation by Rwanda Investigation Bureau (RIB) engineering and legal teams
- Security assessment by the National Cyber Security Authority (NCSA)
- Regulatory compliance review by RURA and BNR
- System integration planning by Rwandan Mobile Network Operators (MNOs)
- Engineering implementation by the iNgabo development team
- Third-party security auditors conducting platform assessments

1.2 Scope

This document covers the complete iNgabo platform technical design including: backend microservices architecture, frontend and mobile client design, AI/ML subsystems, CAMARA API integration, telecom protocol integration, database design (PostgreSQL, Neo4j, Redis, ClickHouse), event streaming architecture (Apache Kafka), security architecture (Zero Trust, RBAC/ABAC, encryption, audit), deployment architecture (Kubernetes, Docker), and observability (Prometheus, Grafana, OpenTelemetry).

Voice analysis capabilities are described at the architecture level; actual voice content processing deployment is conditional on applicable legal authorization, RURA regulatory clearance, and MNO technical capability confirmation. This document clearly distinguishes between currently implemented capabilities, planned integrations, and future roadmap items throughout.

1.3 Relationship to SRS

This TAD is a companion document to the iNgabo Software Requirements Specification v4.0 (iNGBO-SRS-v4.0-2026). The SRS defines what the platform does; this TAD defines how it is built. Requirements are referenced by their SRS identifier (e.g., TFI-001) where relevant.

1.4 Intended Audience

Table 3: Document Audience Guide

Audience	Relevant Sections
RIB Technical Evaluation Team	All sections; especially §6 System Architecture, §17 Investigation Management, §22 Security Architecture
RURA Technical Compliance	§11 CAMARA API Integration, §12 Telecom Integration, §20 Data Architecture, §23 Privacy and Compliance
NCSA Security Assessment	§22 Security Architecture, §23 Privacy and Compliance, §24 Deployment Architecture
MNO Integration Teams	§11 CAMARA API Integration, §12 Telecom Integration, §8 Component Design
iNgabo Engineering Team	All sections
Third-Party Auditors	§22 Security Architecture, §20 Data Architecture, §25 Monitoring and Observability
Senior Government Officials	§ES Executive Summary, §2 Problem Statement, §3 Solution Overview, §30 Future Enhancements

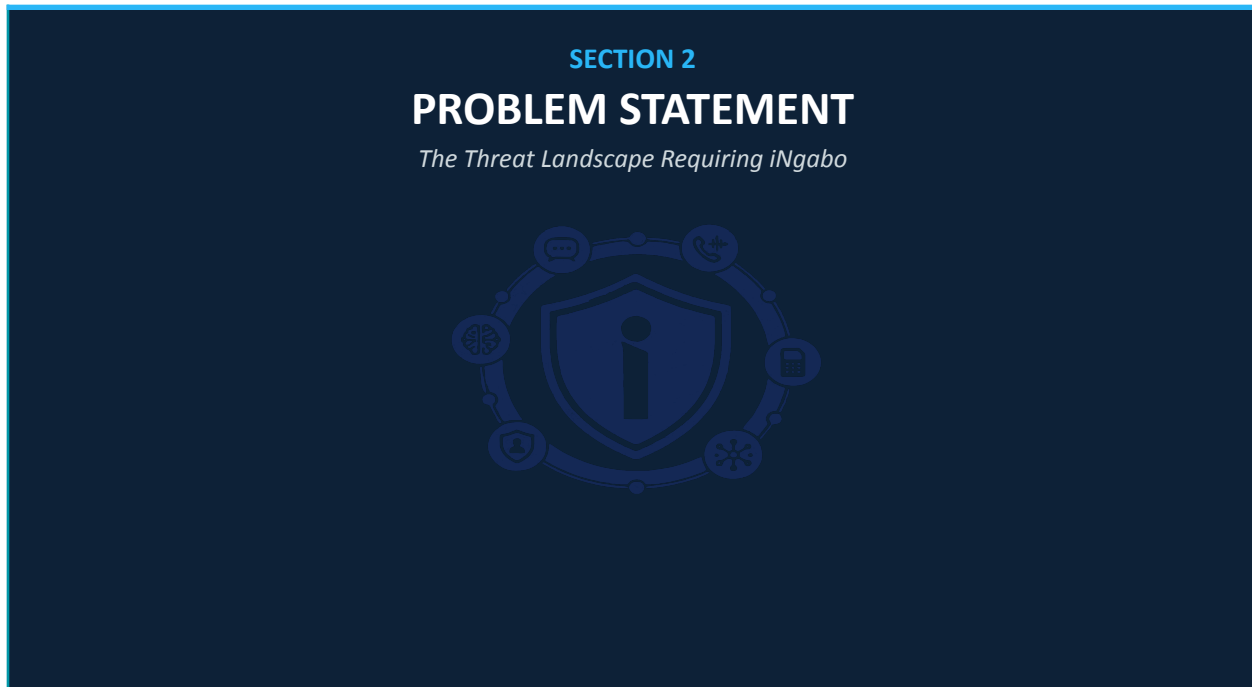
1.5 Definitions and Abbreviations

Table 4: Definitions and Abbreviations

Term	Definition
CAMARA	Cloud and Network Automation API - GSMA/Linux Foundation open network API initiative
CDR	Call Detail Record - telecom call/data session metadata (not content)
IMEI	International Mobile Equipment Identity - unique 15-digit device identifier
MSISDN	Mobile Subscriber ISDN Number - the mobile phone number
HLR	Home Location Register - telecom database storing subscriber information
VLR	Visitor Location Register - temporary subscriber location database
SS7	Signaling System No. 7 - legacy telecom inter-network signaling protocol
DIAMETER	IETF RFC 6733 - successor signaling protocol to SS7 in 4G/5G networks
ZTA	Zero Trust Architecture - NIST SP 800-207 security model
XAI	Explainable AI - AI models that produce human-interpretable decision rationale

Term	Definition
HITL	Human-in-the-Loop - mandatory human review before AI-driven enforcement actions
GNN	Graph Neural Network - ML model operating on graph-structured data
RBAC	Role-Based Access Control
ABAC	Attribute-Based Access Control
mTLS	Mutual Transport Layer Security - bidirectional certificate authentication
STIX	Structured Threat Information eXpression - threat intelligence interchange format
TAXII	Trusted Automated eXchange of Intelligence Information - STIX transport protocol
MNO	Mobile Network Operator (MTN Rwanda, Airtel Rwanda, Liquid Telecom)
RIB	Rwanda Investigation Bureau
RNP	Rwanda National Police
NCSA	National Cyber Security Authority
RURA	Rwanda Utilities Regulatory Authority
BNR	Banque Nationale du Rwanda
K8s	Kubernetes - container orchestration platform
OPA	Open Policy Agent - declarative policy enforcement engine
SPIFFE	Secure Production Identity Framework for Everyone - workload identity
HSM	Hardware Security Module - dedicated cryptographic processing device

2. PROBLEM STATEMENT



2.1 National Fraud Threat Landscape

Rwanda's rapidly expanding digital economy - encompassing mobile money, e-government services, digital banking, and mobile commerce - has created an increasingly attractive target surface for organized fraud networks. These networks operate at speed, scale, and sophistication that outpaces the capabilities of current fragmented, reactive, per-institution security controls.

2.2 Phone Theft and Device Fraud

Mobile device theft in Rwanda involves organized networks that rapidly modify or clone stolen device IMEIs to evade blacklisting. Current recovery mechanisms lack the technical capability to track device network activity post-theft, detect SIM card replacement events that follow theft, or correlate device activity across operators. Law enforcement lacks structured digital evidence pipelines for device-related crimes.

TECHNICAL GAP

Current IMEI blacklisting operates with 24–72 hour propagation delays across operators. During this window, stolen devices are used for fraud, cloned, or exported. No cross-operator, real-time device intelligence exists in Rwanda.

2.3 SIM Swap and Identity Fraud

SIM swap attacks - where a fraudster persuades an MNO to transfer a victim's number to a new SIM - represent the single highest-impact fraud vector in mobile money ecosystems. An attacker with a victim's phone number can intercept OTPs, bypass MFA, and drain mobile money and bank accounts within minutes. Currently, no cross-sector SIM swap intelligence exists; each MNO evaluates swap requests in isolation without visibility into whether the same MSISDN has been attacked on a competitor network.

2.4 SMS and Voice Fraud Campaigns

Smishing and vishing campaigns targeting Rwandan citizens have grown significantly in sophistication. Campaigns now impersonate BNR, RURA, commercial banks, MNOs, and government agencies with high visual and linguistic credibility. Current MNO-level SMS filtering operates on basic keyword rules with no AI-based content analysis, no campaign correlation across operators, and no citizen feedback loop.

2.5 Investigation Infrastructure Gap

Rwanda Investigation Bureau and Rwanda National Police cybercrime units currently lack:

- A unified digital case management system for telecom-related crime
- Real-time device location intelligence tools within legal frameworks
- Fraud network visualization tools to reveal criminal organization structure
- Automated evidence packaging with chain-of-custody compliance
- Cross-case intelligence correlation to identify linked fraud incidents
- AI-assisted investigation recommendations based on case pattern analysis

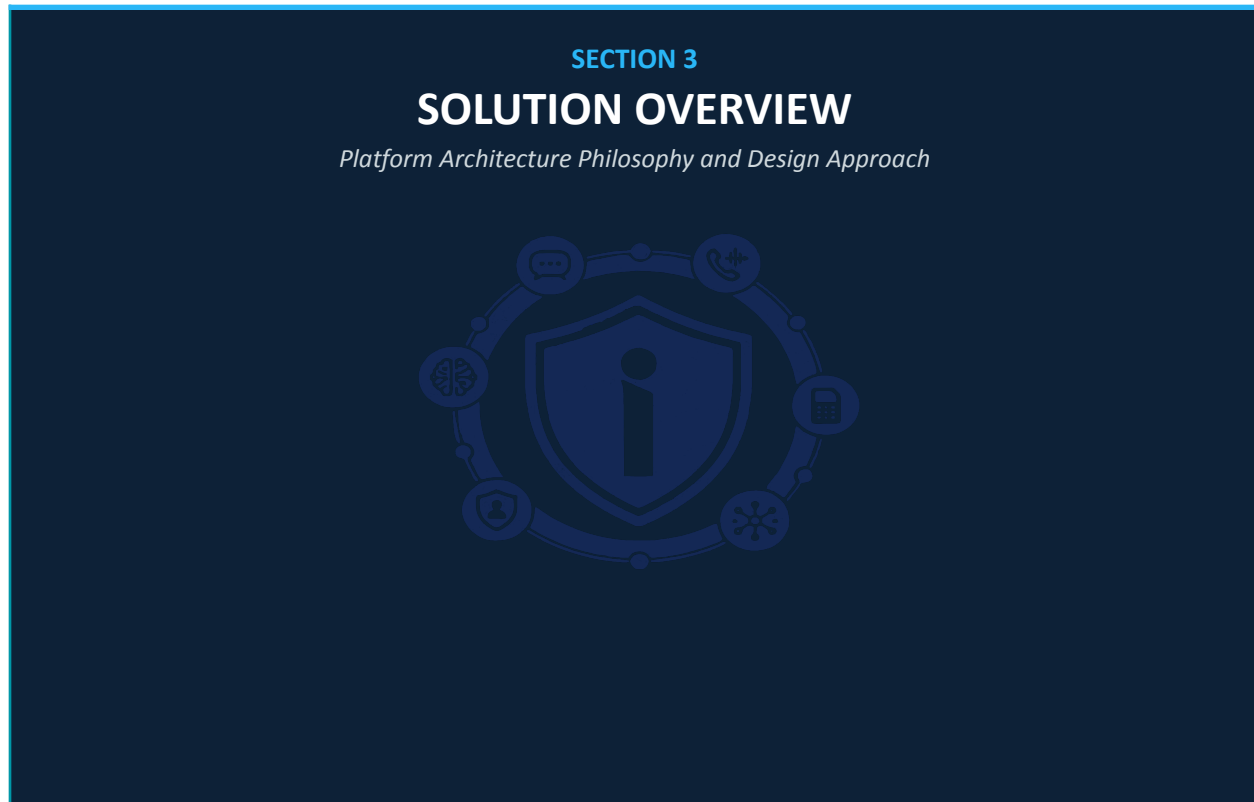
2.6 Structural Failure: Fragmented Defense

Table 5: Defense Gap Analysis

Defense Gap	Current State	iNgabo Target State
Cross-operator intelligence	None - each MNO operates in isolation	Unified cross-operator fraud signal graph
Citizen fraud detection	No proactive citizen alert capability	Real-time SMS/app alerts to at-risk citizens
SIM swap protection	Post-event detection only (account already compromised)	Pre-execution risk scoring and intercept capability
Investigation tooling	Manual, spreadsheet-based case management	AI-assisted digital case management with evidence graph
Device recovery	IMEI blacklist only - no network intelligence	Network-assisted location approximation and activity monitoring & Blacklisting

Defense Gap	Current State	iNgabo Target State
Fraud pattern library	No shared national fraud pattern database	National fraud intelligence hub with 1M+ indicators
Cross-border intelligence	No EAC-level intelligence sharing	Federated EAC fraud intelligence exchange (Phase 5)

3. SOLUTION OVERVIEW



3.1 Platform Design Philosophy

iNgabo is architected on four foundational design principles that distinguish it from conventional enterprise security platforms:

3.1.1 Intelligence Over Data

iNgabo does not create a centralized repository of raw personal data. The platform ingests signals - behavioral patterns, anomaly indicators, risk scores, relationship graphs - and distributes intelligence. Personal data remains with originating institutions under their data governance frameworks. This design is both architecturally superior (no single breach exposes all data) and regulatory-compliant with Rwanda Law No. 058/2021.

3.1.2 Proactive Protection

The architecture prioritizes prevention over detection. Real-time stream processing on all data channels enables the platform to act within the transaction window - before a fraudulent SIM swap completes, before an OTP-based attack succeeds, before a smishing URL is clicked, before a scammer steals citizen's money. This requires sub-second processing pipelines throughout the core fraud detection path.

3.1.3 Zero Trust by Default

No component trusts any other component implicitly. Every service-to-service call carries a cryptographically verified identity, a scoped authorization token, and is validated against runtime policy. This applies equally to internal microservices, external API integrations, and administrative access - no exceptions.

3.1.4 Explainable, Accountable AI

Every AI-generated risk score, fraud classification, or investigation recommendation is accompanied by a human-readable explanation of the contributing signals. No enforcement action (account flag, SIM swap block, LEA referral) is executed purely on AI output without human review. This is not a policy constraint - it is an architectural requirement enforced at the application layer.

3.2 Platform Component Map



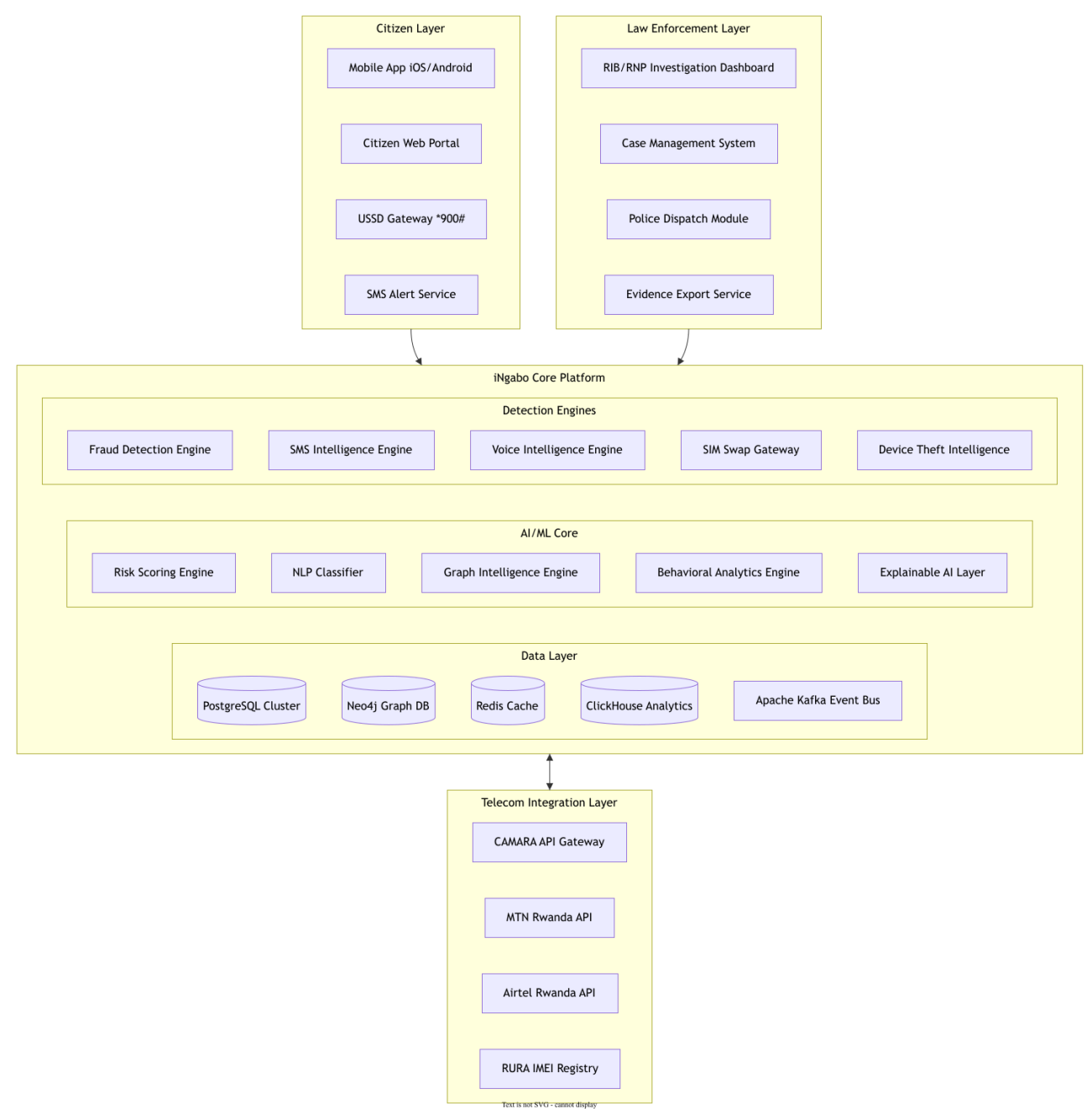


Figure 1: iNgabo High-Level Platform Component Map

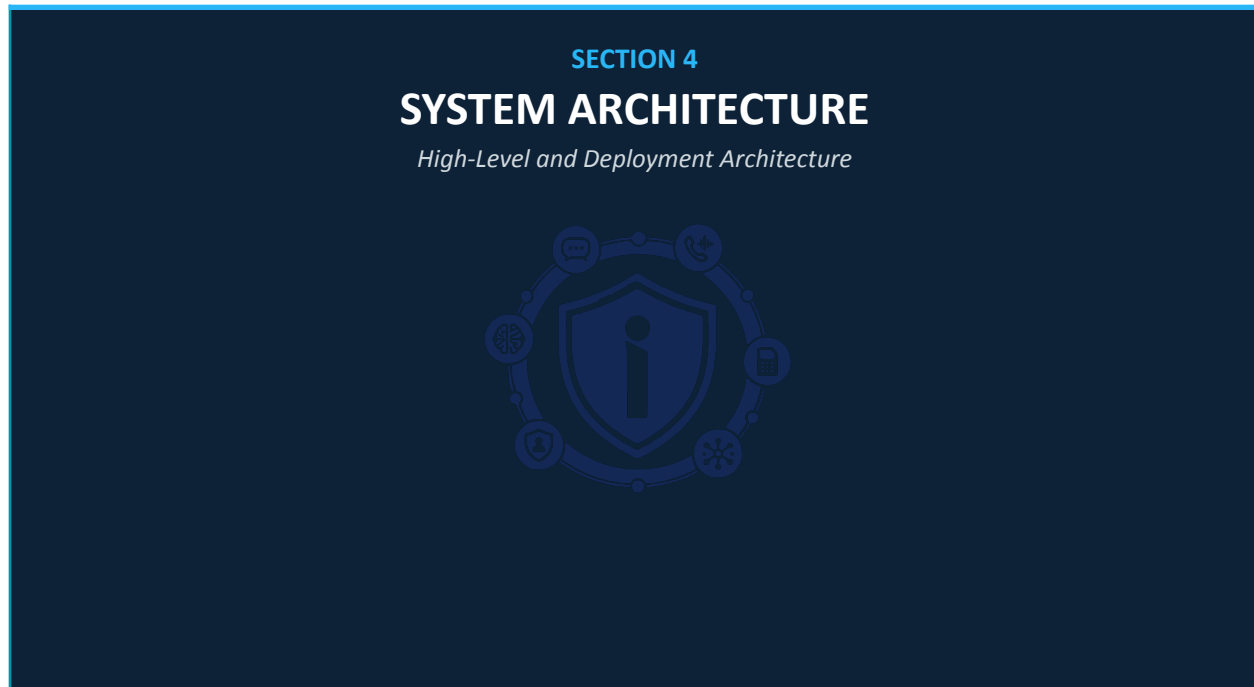
3.3 Data Flow Overview

iNgabo processes four classes of data flows, each with distinct latency requirements, security controls, and processing pathways:

Table 6: iNgabo Data Flow Classes

Flow Class	Latency Req.	Volume	Processing Path	Security Tier
Real-time fraud prevention	< 2 seconds	50K events/sec (peak)	Kafka → Stream Processor → ML Inference → Alert	Critical — HSM-encrypted, mTLS throughout
SIM swap intercept	< 1 second	Variable — event-driven	MNO webhook → SSG → Risk Score → Block/Allow/Review	Critical — synchronous, signed payload
Batch behavioral analysis	< 4 hours (nightly)	Full subscriber baseline	Spark batch → ML training update → Profile store	High — encrypted at rest, access-controlled
Investigation intelligence	< 4 seconds (interactive)	On-demand query	Neo4j graph query → AI recommendation → Dashboard	High — RBAC, full audit trail

4. SYSTEM ARCHITECTURE



4.1 Architecture Overview

iNgabo employs a layered microservices architecture deployed on Kubernetes across multiple availability zones. The architecture separates concerns across six horizontal layers, each with explicit interface contracts, independent scaling, and isolated security domains.

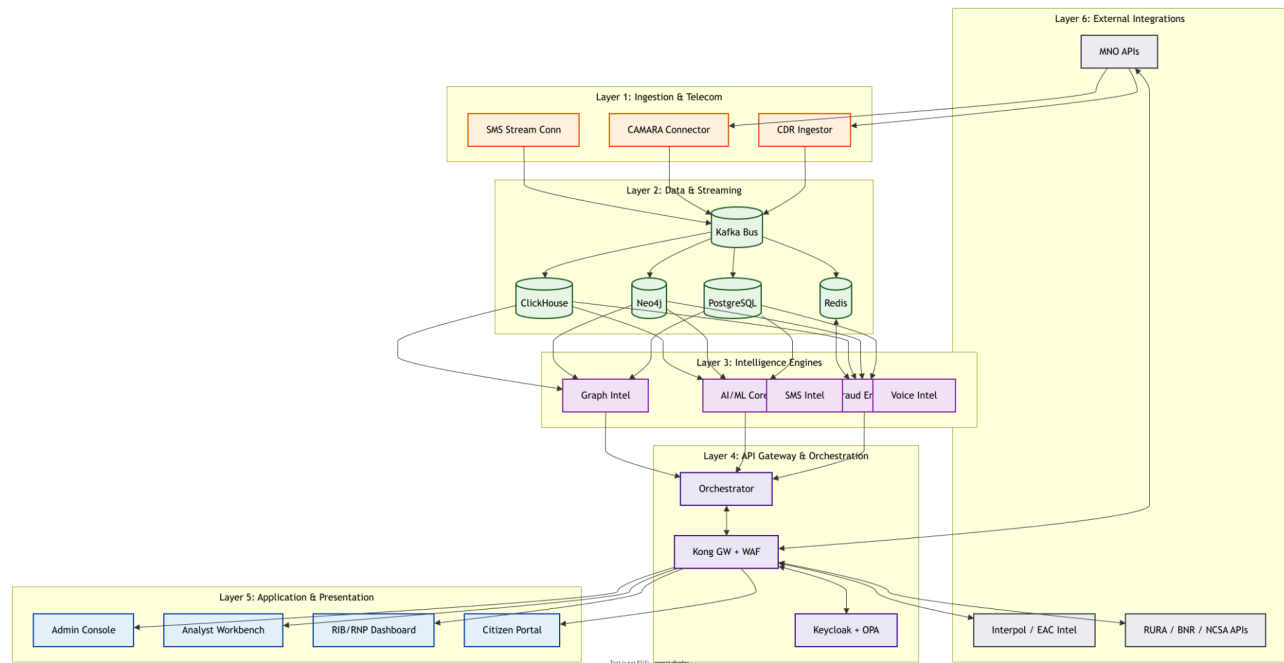


Figure 2: iNgabo Six-Layer Architecture Stack

4.2 Deployment Architecture

The platform is deployed on a Kubernetes cluster spanning three availability zones within Rwanda's data center infrastructure (or a nearest-region cloud provider with Rwanda data residency guarantees). The deployment uses GitOps with ArgoCD for all production changes, ensuring every deployment is auditable and reproducible.

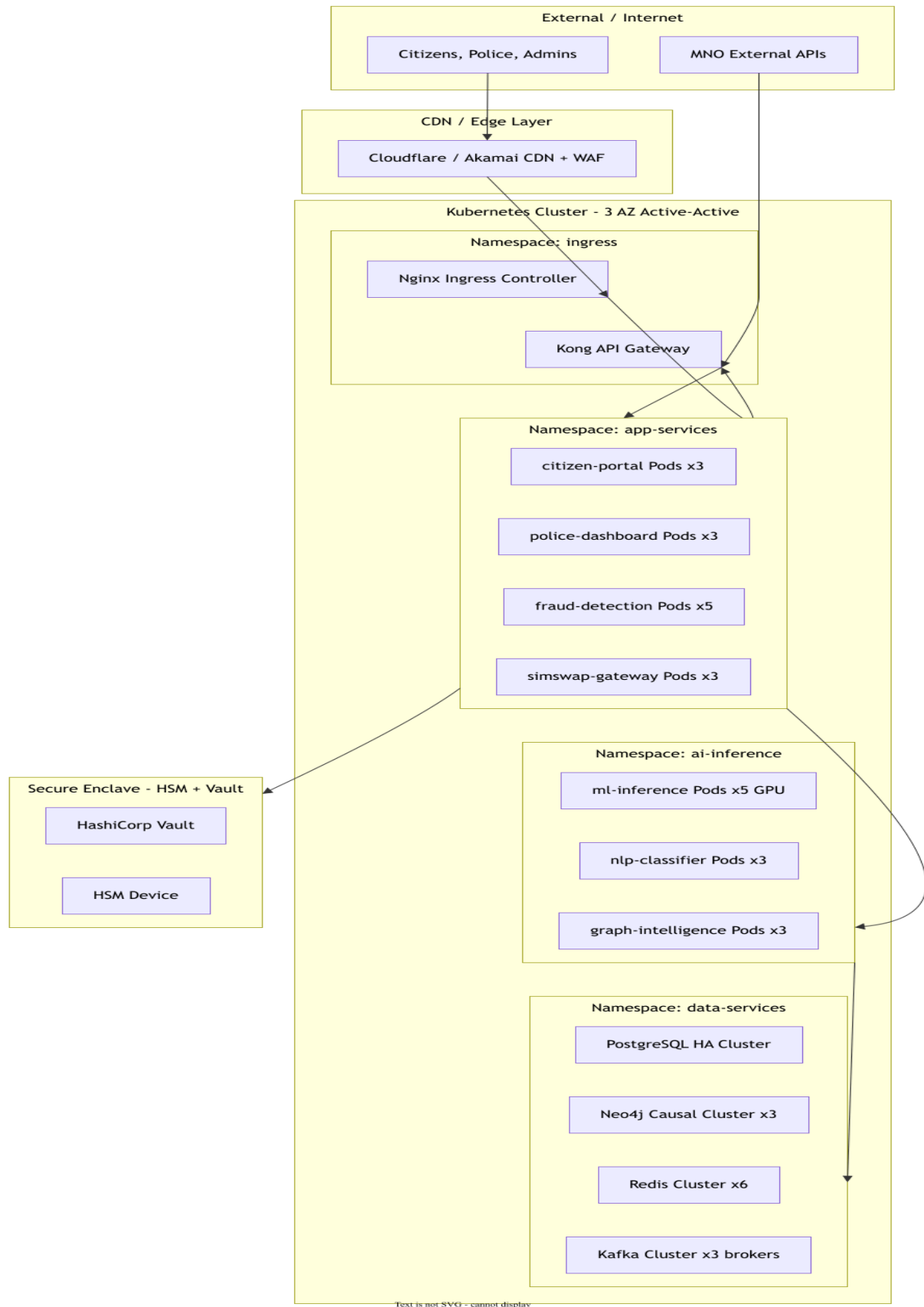


Figure 3: iNgabo Kubernetes Deployment Topology

4.3 Multi-Availability-Zone Resilience

Table 7: Component Resilience Configuration

Component	Deployment Mode	Failover Time	Data Loss Tolerance
Fraud Detection Engine	Active-Active x3 AZ	< 30 seconds (automatic)	Zero — event stream replayed
SIM Swap Gateway	Active-Active x3 AZ	< 10 seconds	Zero — synchronous intercept
PostgreSQL	Primary + 2 synchronous replicas	< 60 seconds (auto-promote)	Zero — synchronous replication
Neo4j Graph DB	Causal cluster (1 Leader + 2 Followers)	< 30 seconds (new leader election)	Zero — Raft consensus
Redis Cache	Cluster mode x6 nodes (3 shards)	< 5 seconds	Acceptable — cache is rebuilt
Kafka Event Bus	3 brokers, replication factor 3	< 30 seconds	Zero — retained on disk
API Gateway (Kong)	x3 replicas per AZ	< 5 seconds	Stateless — instant recovery
Citizen Portal	x3 pods per AZ, CDN-cached	< 2 seconds	Stateless — CDN serves cached

4.4 Network Architecture

All inter-service communication is encrypted with mutual TLS via the Istio service mesh. The network is segmented into security zones with explicit egress controls enforced at the namespace level via Kubernetes NetworkPolicy. No service may communicate with another service outside its authorized dependency graph - this is enforced at the infrastructure level, not the application level.

Table 8: Network Zone Architecture

Network Zone	Services	Ingress Allowed From	Egress Allowed To
DMZ / Edge	Kong API Gateway, Nginx Ingress	Internet (HTTPS:443 only)	App Services Zone only
App Services	Citizen Portal, Police Dashboard, Fraud APIs	DMZ Zone only	AI Zone, Data Zone, Vault
AI/Inference	ML Inference, NLP, Graph Intelligence	App Services Zone only	Data Zone only
Data Services	PostgreSQL, Neo4j, Redis, Kafka, ClickHouse	App Zone, AI Zone only	None (no external egress)

Network Zone	Services	Ingress Allowed From	Egress Allowed To
Secure Enclave	Vault, HSM	App Zone (specific services only)	None (air-gapped from internet)
Telecom Integration	CAMARA Gateway, CDR Ingestors	Telecom DMZ (MNO peering)	Data Zone only
Telecom DMZ	MNO VPN Termination	MNO dedicated circuits only	Telecom Integration Zone

5. TECHNICAL ARCHITECTURE



5.1 Event-Driven Architecture

iNgabo's core processing pipeline is event-driven, built on Apache Kafka as the central event bus. All data ingestion, fraud signal processing, alert dispatching, and audit events flow through Kafka topics with defined schemas (Apache Avro + Schema Registry), ensuring strong backward compatibility and schema evolution management.

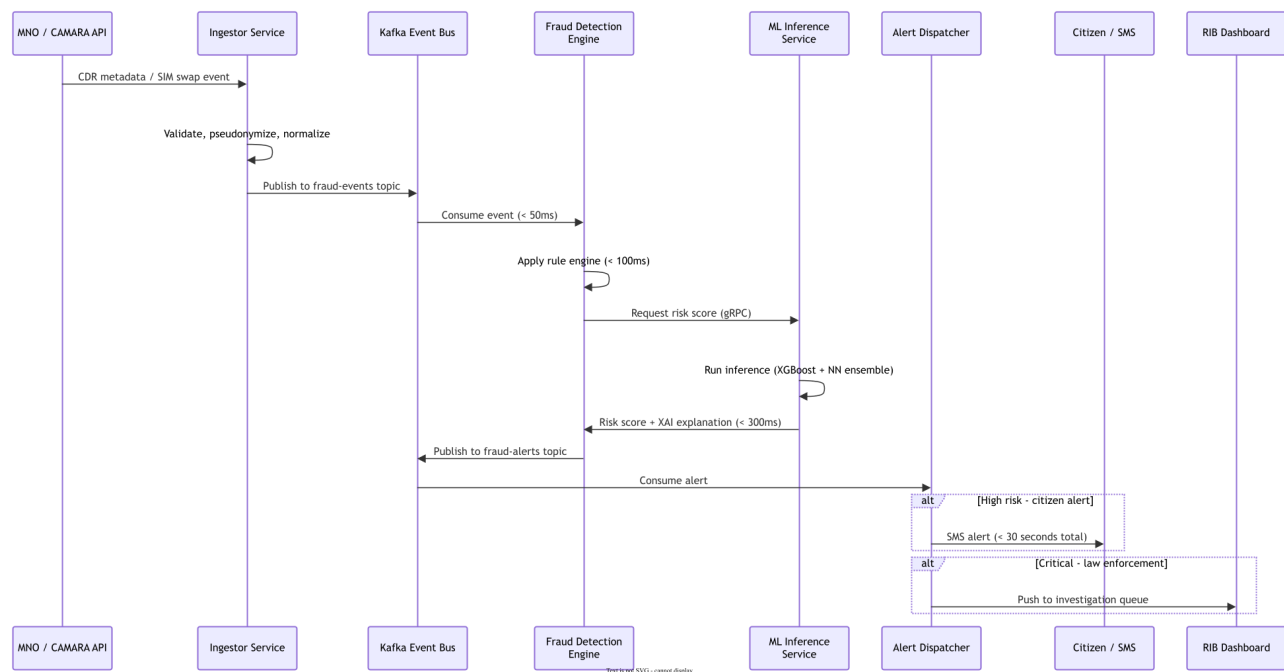


Figure 4: Core Fraud Detection Event Flow — Sequence Diagram

5.2 API Architecture

All external and inter-service APIs are versioned REST or gRPC. External APIs are exposed through Kong API Gateway with rate limiting, authentication, and WAF inspection. Internal microservice communication uses gRPC for synchronous calls (risk scoring, SIM swap intercept) and Kafka for asynchronous event flows.

Table 9: API Architecture Summary

API Category	Protocol	Authentication	Rate Limiting	Use Case
Public Citizen APIs	REST HTTPS	API Key (read-only)	60 req/min/IP	Number verification, scam check, fraud report
Partner Institution APIs	REST HTTPS + mTLS	OAuth2 Client Credentials + mTLS	10,000 req/min	Risk scores, SIM swap check, IMEI status
MNO Integration APIs	REST HTTPS + mTLS	mTLS + API Key (MNO-specific)	Unlimited (trusted)	CDR ingest, SIM swap webhook, CAMARA calls
Law Enforcement APIs	REST HTTPS + mTLS	mTLS + JWT (RIB PKI)	1,000 req/min	Case management, intelligence queries, evidence export
Internal Microservice (sync)	gRPC + mTLS	SPIFFE/SPIRE workload identity	None (internal mesh)	ML inference, SIM swap scoring, identity check
Internal Microservice (async)	Kafka + mTLS	SASL/SCRAM + TLS	None (internal bus)	Fraud events, alerts, audit log events
Administrative APIs	REST HTTPS + mTLS	mTLS + MFA-backed JWT	100 req/min (per admin)	Platform configuration, user management, audit review

5.3 Service Mesh - Istio

The Istio service mesh provides transparent mutual TLS for all pod-to-pod communication, traffic management (circuit breaking, retry policies, load balancing), and distributed tracing integration. Every microservice is automatically enrolled in the mesh via sidecar injection - no application code changes required for baseline security.

5.3.1 Istio Traffic Policies

- PeerAuthentication: STRICT mode cluster-wide - all pods must present valid mTLS certificate
- AuthorizationPolicy: per-service policies defining exactly which services may call which endpoints
- DestinationRule: connection pool limits, outlier detection, and TLS mode enforcement
- VirtualService: traffic weighting for canary deployments and A/B testing of ML models

5.4 Configuration and Secrets Management

HashiCorp Vault manages all platform secrets - database credentials, API keys, encryption keys, and TLS certificates. Vault is backed by a Hardware Security Module (HSM) for root key protection. Application pods retrieve secrets at startup via the Vault Agent Injector sidecar, with automatic lease renewal and key rotation without pod restarts.

Table 10: Secrets Management Configuration

Secret Type	Storage	Rotation Policy	Access Control
Database credentials	Vault Dynamic Secrets	Auto-rotated every 1 hour	Per-service policy - only owning service
API keys (MNO/Partner)	Vault KV v2	Manual rotation on compromise; scheduled 90-day	API team role only
mTLS certificates	Vault PKI + cert-manager	Auto-renewed 24h before expiry	Istio / cert-manager only
Encryption keys (AES-256)	Vault Transit Engine (HSM-backed)	Annual scheduled rotation	Encryption service only - no direct key access
HSM root key	HSM hardware (m-of-n quorum)	Annual with quorum ceremony	2-of-3 custodian quorum required
JWT signing keys	Vault PKI	Rotated every 7 days	Keycloak only
Kafka SASL credentials	Vault Dynamic Secrets	Auto-rotated every 4 hours	Per-consumer-group policy

6. MICROSERVICES ARCHITECTURE



6.1 Service Decomposition Principles

Services are decomposed along domain boundaries derived from the iNgabo operational model. Each service owns its data store, exposes a versioned API contract, and can be deployed, scaled, and updated independently. Services are sized to minimize inter-service chattiness while maintaining clear domain separation.

6.2 Service Catalog

Table 11: iNgabo Microservice Catalog

Service Name	Domain	Technology	Responsibilities	Scales On
cdr-ingestor	Ingestion	NestJS, Kafka	Ingest MNO CDR metadata streams; validate schema; pseudonymize MSISDNs; publish to fraud-events topic	CDR volume
simswap-gateway	SIM Swap	NestJS, PostgreSQL, Redis	Intercept SIM swap webhooks from MNOs; compute risk score; return block/allow/escalate decision within 1s	SIM swap event rate
fraud-detection-engine	Detection	Python, FastAPI, Kafka	Consume fraud events; apply rule engine; invoke ML inference; publish fraud alerts	Event throughput
ml-inference-service	AI/ML	Python, FastAPI, TensorFlow Serving	Serve trained ML models; compute risk scores; invoke XAI engine; return score + explanation	Inference request rate (GPU)
nlp-classifier	AI/ML NLP	Python, FastAPI, HuggingFace	Classify SMS/message content; detect phishing URLs; score	SMS classification

Service Name	Domain	Technology	Responsibilities	Scales On
			scam probability; support Kinyarwanda	ion volume
graph-intelligence-engine	AI/ML Graph	Python, FastAPI, Neo4j	Execute GNN fraud network analysis; community detection; path analysis; betweenness centrality	Query complexity
behavioral-analytics	AI/ML	Python, FastAPI, TimescaleDB	Maintain behavioral baselines; detect statistical anomalies; score behavioral risk	Account baseline count
voice-intelligence-engine	Voice	Python, FastAPI	Analyze voice metadata; vishing pattern detection; caller reputation scoring (no content recording)	Call metadata volume
device-theft-intelligence	Device	NestJS, PostgreSQL, Redis	IMEI registry integration; stolen device tracking; SIM-IMEI pairing anomaly detection; recovery workflow	Device event rate
sms-intelligence-engine	SMS	NestJS, Kafka, NLP-svc	Consume SMS metadata; invoke NLP classifier; detect campaigns; dispatch alerts	SMS volume
alert-dispatcher	Alerting	NestJS, Redis	Consume fraud alerts; route to correct channel (SMS/app/dashboard); deduplicate; throttle per-citizen	Alert volume
citizen-api	Citizen	NestJS, PostgreSQL	Citizen portal backend; fraud reporting; number verification; case tracking; USSD bridge	User request rate
police-dashboard-api	LEA	NestJS, PostgreSQL, Neo4j	RIB/RNP case management; investigation workflow; evidence export; AI recommendations	Investigator sessions
camara-connector	Integration	NestJS	CAMARA API abstraction; Device Location; SIM Swap; Number Verification; KYC Match, Device status, Roaming, Location retrieval.	CAMARA call volume
threat-intelligence-hub	Intelligence	NestJS, PostgreSQL	IOC management; STIX/TAXII feeds; inter-institution sharing; intelligence products	Indicator count
regulatory-reporting	Compliance	NestJS	Automated RURA/BNR/NCSA report generation; audit log extraction; compliance evidence	Report schedule
auth-service	Security	Keycloak, NestJS, OPA	OIDC/OAuth2 identity provider; RBAC/ABAC policy enforcement; session management; MFA	Auth request rate

Service Name	Domain	Technology	Responsibilities	Scales On
audit-log-service	Security	NestJS, ClickHouse	Immutable audit event consumer; tamper-evident storage; audit query API	Event write rate
notification-service	Communication	NestJS	SMS gateway integration (Africa's Talking); push notification; email alerts	Notification volume
ussd-gateway-service	Communication	NestJS	USSD session management; citizen menu navigation; Africa's Talking integration	USSD session count

6.3 Service Interaction Diagram

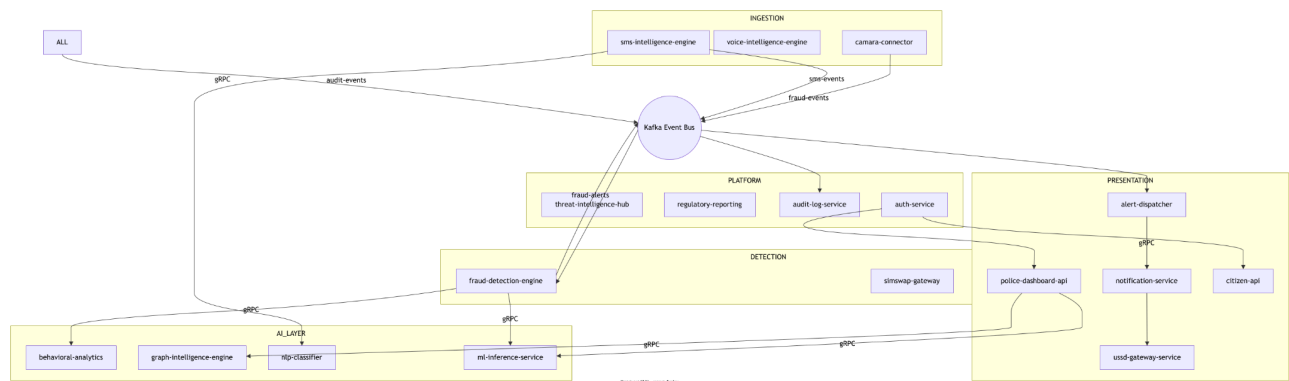


Figure 5: Microservice Interaction Map — Core Processing Flows

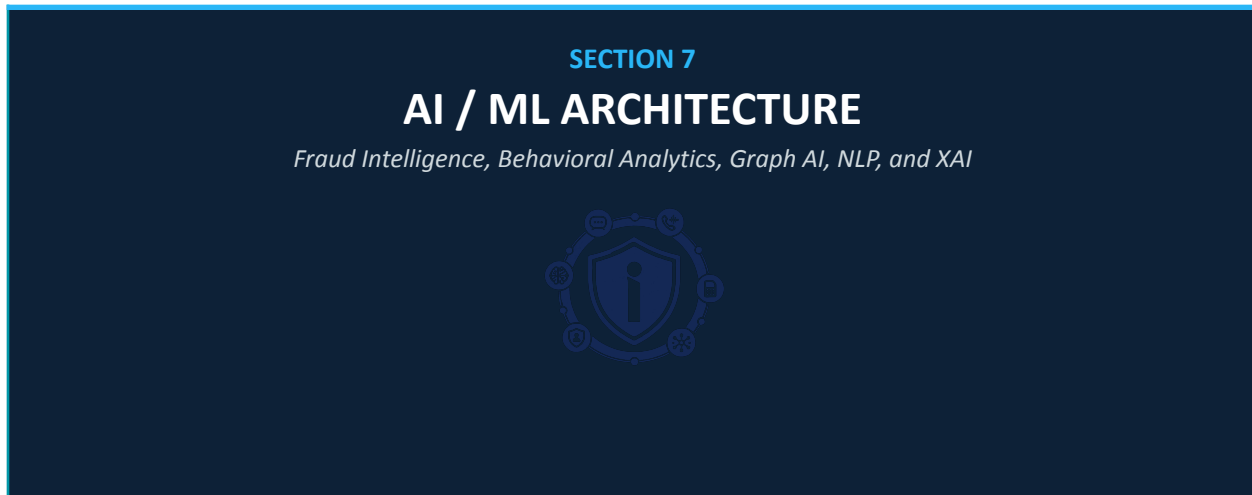
6.4 Data Ownership

Table 12: Service Data Ownership Matrix

Service	Primary Data Store	Shared Read Access	Data Class
fraud-detection-engine	ClickHouse (fraud events)	threat-intelligence-hub, police-dashboard-api	Internal / Confidential
simswap-gateway	PostgreSQL (swap events, risk scores)	police-dashboard-api, regulatory-reporting	Restricted
behavioral-analytics	TimescaleDB (behavioral profiles)	ml-inference-service (read-only)	Confidential
device-theft-intelligence	PostgreSQL (device registry, cases)	police-dashboard-api, citizen-api	Restricted / Internal
citizen-api	PostgreSQL (citizen accounts, reports)	alert-dispatcher (read-only)	Personal / Confidential

Service	Primary Data Store	Shared Read Access	Data Class
police-dashboard-api	PostgreSQL (cases, evidence), Neo4j (graph)	regulatory-reporting, audit-log-service	Restricted
graph-intelligence-engine	Neo4j (full platform graph)	police-dashboard-api, threat-intelligence-hub	Restricted
audit-log-service	ClickHouse (write-once audit log)	regulatory-reporting (read-only)	Restricted — immutable
threat-intelligence-hub	PostgreSQL (IOCs, indicators)	All services (read via API only)	Confidential

7. AI AND MACHINE LEARNING ARCHITECTURE



7.1 AI Architecture Overview

The iNgabo AI/ML core is a multi-model ensemble system combining four complementary analytical approaches: supervised fraud classification, unsupervised behavioral anomaly detection, graph neural network analysis, and NLP-based content classification. Each model addresses distinct aspects of the fraud landscape; the ensemble fusion layer combines their outputs into a unified risk score with cross-signal explainability.

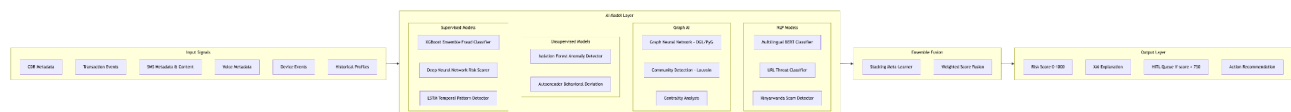


Figure 6: iNgabo AI/ML Architecture - Model Ensemble Design

7.2 Fraud Risk Scoring Engine

7.2.1 Model Architecture

The fraud risk scoring engine uses a two-stage ensemble: a first-stage layer of specialized models (XGBoost for tabular CDR features, LSTM for temporal patterns, GNN for graph features), followed by a meta-learner (gradient boosting stacker) that combines first-stage outputs with contextual features into the final 0–1000 risk score.

Table 13: Fraud Risk Scoring Model Components

Model Component	Type	Input Features	Output	Latency Budget
XGBoost Fraud Classifier	Supervised — Gradient Boosting	CDR features: velocity, geography, device, time-of-day, PRS flags, roaming status	P(fraud): 0.0–1.0 + feature importance	< 50ms
LSTM Temporal Detector	Supervised — Recurrent NN	Time-series of user behavior over 30-day window: call patterns, transaction timing, location sequence	Temporal anomaly score 0–1	< 80ms
Deep Neural Network	Supervised — Feed-Forward	High-cardinality entity embeddings: MSISDN, IMEI, cell tower, counterparty embeddings	Entity-contextualized fraud probability	< 60ms
Isolation Forest	Unsupervised — Ensemble	Real-time feature vector vs. behavioral baseline (200 features)	Anomaly score — distance from baseline	< 30ms
GNN (graph features)	Semi-supervised — Graph NN	Local subgraph: 2-hop neighborhood of target entity in Neo4j	Graph-contextualized risk embedding	< 150ms
Meta-Learner Stacker	Supervised — GBM	First-stage outputs + entity metadata + historical fraud rate	Final risk score 0–1000	< 40ms

7.2.2 Feature Engineering

Feature engineering operates in two modes: real-time features computed within the event processing pipeline (< 100ms budget), and pre-computed behavioral profile features updated nightly in batch.

Real-Time Features (computed at scoring time)

- Call velocity: number of unique destinations in last 1h, 6h, 24h
- Geographic velocity: distance between consecutive cell tower registrations vs. time elapsed
- SIM age: days since current SIM registration (low SIM age + high-value transaction = elevated risk)
- Device-SIM pairing: number of SIMs used by this IMEI in last 30 days
- Counterparty risk: fraud risk score of transaction counterparties
- Time-of-day anomaly: deviation from historical activity time distribution
- International CLI mismatch: discrepancy between declared and network-derived origin

Pre-Computed Behavioral Profile Features (nightly batch)

- 30-day average call volume, duration distribution, geographic spread
- Historical transaction amount distribution (mean, std dev, max)
- Typical active hours (hour-of-day histogram)
- Network neighborhood stability (how often counterparties change)

- Historical SIM swap frequency
- Prior fraud flag history (own account and counterparty network)

7.3 Behavioral Analytics Engine

The behavioral analytics engine maintains per-entity (MSISDN, IMEI, account) behavioral baselines using streaming statistics updated continuously. Statistical process control techniques (CUSUM, EWMA) detect shifts in baseline behavior that precede fraud events - enabling detection of account compromise before the fraudulent transaction itself.

IMPLEMENTATION NOTE - Phase Dependency

Full behavioral baseline requires minimum 30 days of live data per entity. In Phase 2 (initial telecom integration), baselines are pre-seeded with synthetic-data priors and updated as live data accumulates. Anomaly detection sensitivity is reduced for entities with insufficient baseline history; the risk scoring engine weights behavioral features accordingly.

7.4 Graph Intelligence Engine

7.4.1 Neo4j Graph Model

The graph intelligence engine operates on a property graph stored in Neo4j. Every entity in the fraud ecosystem is a node; every meaningful relationship is an edge. The power of the graph model is its ability to reveal organized criminal networks invisible to per-transaction analysis.

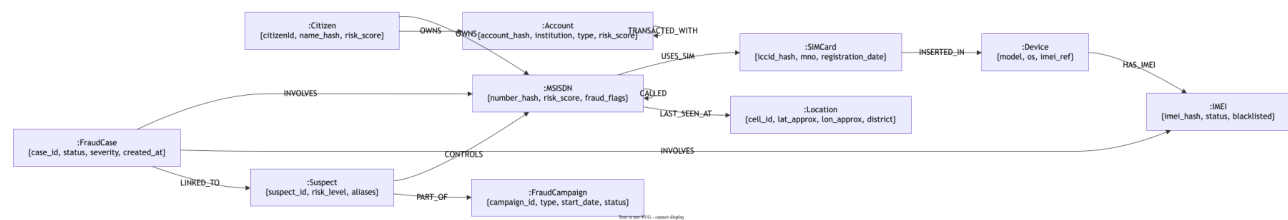


Figure 7: Neo4j Property Graph Schema - iNgabo Fraud Intelligence Graph

7.4.2 Graph Algorithms Applied

Table 14: Graph Algorithms in iNgabo Fraud Intelligence

Algorithm	Library	Fraud Pattern Detected	Typical Result
Louvain Community Detection	Neo4j GDS	Fraud ring identification — clusters of tightly connected suspicious entities	Reveals groups of 5–200 coordinated accounts/devices

Algorithm	Library	Fraud Pattern Detected	Typical Result
PageRank / ArticleRank	Neo4j GDS	Central coordinator nodes in fraud networks (money mule hubs, campaign orchestrators)	Top-ranked nodes = primary investigation targets
Betweenness Centrality	Neo4j GDS	Bridge nodes connecting otherwise separate fraud clusters	Identifies network brokers and SIM farm operators
Weakly Connected Components	Neo4j GDS	Scope of a fraud network — which entities are in the same criminal ecosystem	Defines case scope for investigation planning
Shortest Path (Dijkstra)	Neo4j GDS / Cypher	Relationship chain between suspect and victim	Evidence chain for LEA prosecution packages
Node Similarity	Neo4j GDS	Accounts/devices with similar behavioral signatures (coordinated bots or mule farms)	Groups similar-behaving entities for campaign attribution
Graph Neural Network (GNN)	DGL + PyTorch Geometric	Learn complex fraud patterns not expressible as simple rules	Inductive detection on new entities based on graph structure
Temporal Graph Analysis	Custom Cypher + ClickHouse	Time-ordered event sequences revealing fraud campaign progression	Campaign timeline reconstruction for investigation

7.5 NLP Engine

7.5.1 Model Stack

The NLP engine uses a fine-tuned multilingual BERT model (mBERT) as the base, with domain-specific fine-tuning on a Rwandan SMS fraud dataset combining synthetic examples, known scam patterns, and operator-contributed anonymized samples. A separate Kinyarwanda-optimized model handles local-language content with higher accuracy than the multilingual base.

Table 15: NLP Engine Model Stack

NLP Component	Base Model	Training Data	Task	Output
SMS Fraud Classifier	mBERT fine-tuned	Rwandan SMS fraud corpus (synthetic + real, anonymized)	Multi-class: Safe / Suspicious / Fraudulent / Critical	Class label + confidence + top-3 signal phrases
URL Threat Classifier	Custom CNN + URLScan.io integration	Known malicious URLs, phishing domains, brand impersonation patterns	Binary: Safe / Malicious + subcategory	Threat label + domain reputation score
Kinyarwanda Scam Detector	KinyaBERT (pre-trained) + fine-tuned	Kinyarwanda scam message corpus	Multi-class fraud classification	Class + confidence + Kinyarwanda

NLP Component	Base Model	Training Data	Task	Output
				explanation for agents
Entity Extraction	spaCy + custom NER model	Annotated SMS corpus	Extract: phone numbers, account numbers, URLs, amounts	Structured entities for graph enrichment
Campaign Fingerprinter	MinHash LSH	Processed SMS feature vectors	Near-duplicate detection across large SMS volumes	Campaign cluster ID + similarity score

7.6 Explainable AI (XAI) Framework

Every fraud risk score produced by iNgabo is accompanied by an XAI explanation generated by the SHAP (SHapley Additive exPlanations) framework. The explanation identifies the top contributing features to the score, their direction of influence, and their magnitude — presented in both technical format (for analyst tools) and plain-language format (for citizen and officer interfaces).

```
// Example XAI output structure (JSON)
{
  "risk_score": 847,
  "risk_level": "CRITICAL",
  "explanation": {
    "summary": "High fraud risk: SIM recently replaced, device linked to 3 previous fraud cases, unusual after-midnight activity spike",
    "top_factors": [
      { "feature": "sim_age_days", "value": 1, "contribution": +182, "direction": "INCREASES_RISK",
        "plain_language": "SIM card is only 1 day old – very recent replacement" },
      { "feature": "device_fraud_history", "value": 3, "contribution": +231, "direction": "INCREASES_RISK",
        "plain_language": "This device (IMEI) has been associated with 3 previous fraud cases" },
      { "feature": "activity_hour_anomaly", "value": 2.8, "contribution": +134, "direction": "INCREASES_RISK",
        "plain_language": "Activity at 02:34 AM is 2.8 standard deviations from this account normal pattern" },
      { "feature": "account_history_months", "value": 18, "contribution": -88, "direction": "REDUCES_RISK",
        "plain_language": "Account has 18 months of clean history" }
    ],
    "hitl_required": true,
    "hitl_reason": "Score exceeds 750 threshold – human review required before enforcement action"
  }
}
```

Figure 8: Example XAI Output - Fraud Risk Score with SHAP Explanation

7.7 Human-in-the-Loop (HITL) Framework

iNgabo enforces mandatory human review before any AI-generated decision results in an enforcement action. The HITL framework defines four escalation tiers based on risk score and action type:

Table 16: HITL Escalation Framework

Risk Score Range	Risk Level	Automated Actions Permitted	HITL Required For
0 – 299	LOW	None- informational only	No HITL required
300 – 499	MEDIUM	Proactive citizen alert; enhanced monitoring flag	Any account restriction action
500 – 749	HIGH	MNO advisory alert; SIM swap enhanced scrutiny flag	Account suspension, LEA referral, blacklisting
750 – 899	CRITICAL	SIM swap temporary hold (max 4 hours); citizen alert	All enforcement actions; mandatory analyst review within 2 hours
900 – 1000	IMMINENT	SIM swap block (max 24 hours); citizen emergency alert	All actions; senior analyst + supervisor sign-off; RIB notification

7.8 Model Lifecycle Management

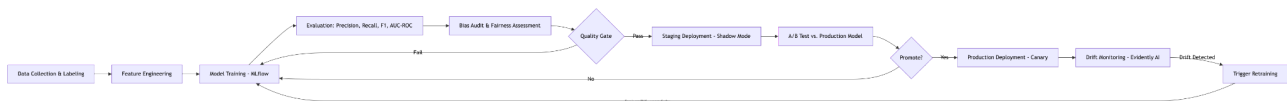


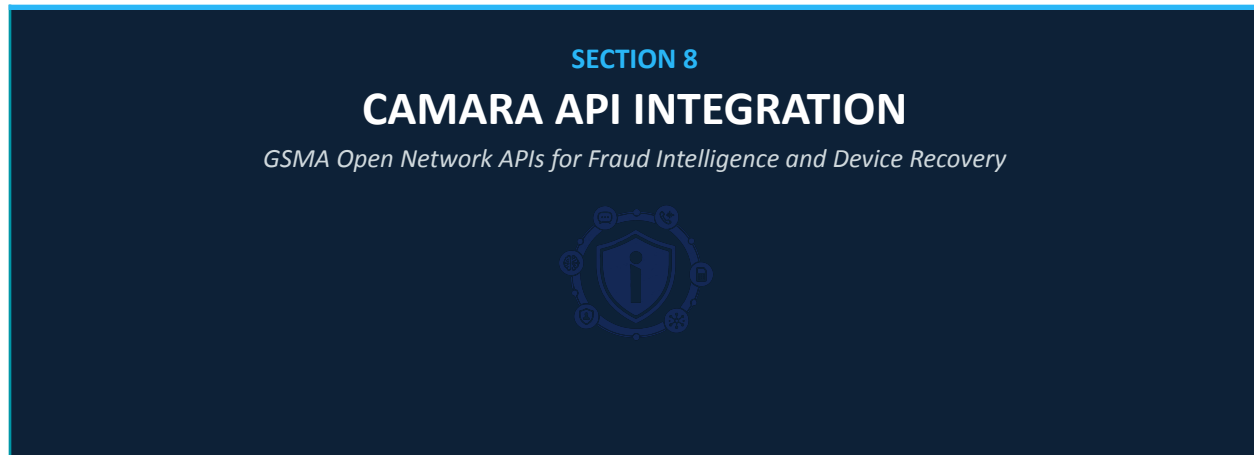
Figure 9: ML Model Lifecycle - Training to Production Pipeline

Table 17: Model Lifecycle Stage Controls

Lifecycle Stage	Tool	Trigger	Quality Gate
Data collection & labeling	Label Studio, RURA-authorized data	Continuous + event-driven	> 95% label agreement (2 annotators)
Feature engineering	Apache Spark, custom Python pipelines	On new training run trigger	Feature completeness > 99%; no data leakage
Model training	MLflow, Kubernetes Job	Weekly scheduled + drift trigger	AUC-ROC > 0.92; Precision > 0.88; Recall > 0.85
Bias audit	Fairlearn, Aequitas	Every training run	No demographic group with > 15% FPR disparity

Lifecycle Stage	Tool	Trigger	Quality Gate
Staging / shadow mode	Istio traffic mirroring (5% traffic)	Post quality gate pass	Shadow performance within 2% of production
Production deployment	ArgoCD canary (10% → 50% → 100%)	Manual approval after staging	P95 latency ≤ production; error rate ≤ 0.1%
Drift monitoring	Evidently AI, custom dashboards	Continuous - daily reports	PSI < 0.2; CSI < 0.1; model performance within 3%
Automated retraining trigger	MLflow + Argo Workflows	Drift detected OR weekly cadence	All quality gates must pass before promotion

8. CAMARA API INTEGRATION



8.1 CAMARA Overview

CAMARA is an open-source project hosted by the Linux Foundation under the GSMA Open Gateway initiative. It defines standardized, RESTful network exposure APIs that allow authorized applications to leverage mobile network capabilities - device location, subscriber verification, SIM swap detection, and more - through a consistent, operator-agnostic interface.

iNgabo integrates CAMARA APIs through a dedicated CAMARA Connector service that abstracts operator-specific implementations, manages API credentials and token lifecycle, enforces rate limits, and logs all API interactions to the immutable audit trail. All CAMARA calls operate under formal Data Sharing Agreements with each MNO, authorized by RURA under the applicable telecom regulation framework.

8.2 CAMARA API Integration Status

Table 18: CAMARA API Integration Roadmap

CAMARA API	Status	Use Case in iNgabo	Regulatory Requirement
SIM Swap	PLANNED — Phase 2	Query when last SIM swap occurred for a MSISDN; high-recency swap = fraud risk signal	RURA Data Sharing Agreement; subscriber consent model
Device Location (Network-Based)	PLANNED — Phase 2/3	Approximate device location for stolen phone recovery; geofencing alerts	RURA + RIB joint authorization; strict legal framework; LEA use only
Number Verification	PLANNED — Phase 2	Verify that a phone number is in possession of the claimed user; anti-impersonation	RURA Data Sharing Agreement; user consent required

CAMARA API	Status	Use Case in iNgabo	Regulatory Requirement
Device Status	PLANNED — Phase 3	Check if device is reachable on network; detect if device has been powered off post-theft	RURA Data Sharing Agreement
KYC Match	PLANNED — Phase 3	Verify subscriber identity attributes against MNO KYC records for account opening fraud prevention	RURA + BNR; subscriber consent; privacy-by-design query
Quality on Demand (QoD)	FUTURE — Phase 5	Priority network session for critical fraud alert delivery in congested conditions	Commercial agreement with MNOs; RURA awareness
Home Devices	FUTURE — Phase 5+	IoT device fraud detection in smart city context	To be determined

8.3 CAMARA Integration Architecture

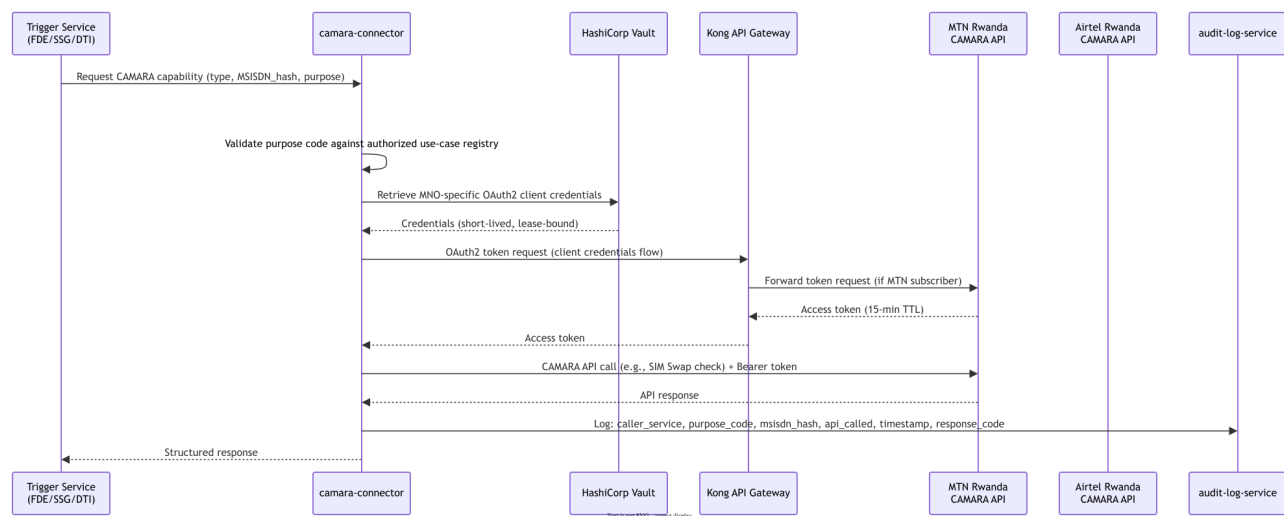


Figure 10: CAMARA API Integration — Authentication and Audit Flow

8.4 Purpose-Based Access Control for CAMARA

All CAMARA API calls are governed by a purpose-based access control model. No service may invoke a CAMARA API without declaring a pre-approved purpose code. The CAMARA Connector validates the calling service's identity and purpose before executing any network API call.

Table 19: CAMARA Purpose-Based Access Control

Purpose Code	Allowed Callers	CAMARA APIs Permitted	Authorization Required
FRAUD_PREVENTION	fraud-detection-engine, simswap-gateway	SIM Swap, Device Status	RURA Data Sharing Agreement
STOLEN_DEVICE_RECOVERY	device-theft-intelligence	Device Location (approximate), Device Status	RIB formal case reference required per call
IDENTITY_VERIFICATION	citizen-api (with user consent)	Number Verification, KYC Match	User explicit consent; logged per GDPR-equiv.
LEA_INVESTIGATION	police-dashboard-api	Device Location (precise — Phase 3), Device Status	RIB court order reference required; NCSA notification
CITIZEN_SELF_CHECK	citizen-api (authenticated citizen)	SIM Swap (own number only), Number Verification	Citizen authenticated session; own MSISDN only

8.5 Privacy Controls for Network API Usage

- MSISDN is never transmitted in plain text to CAMARA APIs - requests use operator-specific subscriber references derived through a one-way function
- Device Location responses are aggregated to district/sector level for citizen-protection use cases; precise location requires RIB court order reference
- All CAMARA API calls produce immutable audit log entries with: requesting service, purpose code, subscriber reference hash, timestamp, response code
- Subscriber notification is sent (SMS) for any Device Location API call within 24 hours of the call, unless RIB has obtained a legal suppression order
- CAMARA API usage statistics are reported to RURA monthly as part of the data sharing agreement compliance obligations

9. TELECOM INTEGRATION ARCHITECTURE

SECTION 9

TELECOM INTEGRATION

MNO Data Ingestion, Signaling Intelligence, and Protocol Coverage



9.1 Integration Model

iNgabo integrates with Rwandan Mobile Network Operators through a layered integration model that respects data minimization principles: the platform receives the minimum data necessary for each fraud detection function, pseudonymized at ingestion, under formal data sharing agreements. Raw subscriber personal data is never ingested - only event metadata, behavioral signals, and fraud risk indicators.

Table 20: Telecom Integration Layer Data Flows

Integration Layer	Data Exchanged	Direction	Protocol	Security
CDR Metadata Stream	Call event metadata: MSISDN hash, IMEI hash, cell ID, timestamp, duration, call type, roaming flag	MNO → iNgabo	Kafka Connect over TLS + mTLS	mTLS + field-level pseudonymization at source
SIM Swap Webhook	Swap request context: MSISDN hash, new ICCID, swap timestamp, channel (app/store/IVR)	MNO → iNgabo (real-time push)	REST HTTPS webhook + mTLS	mTLS + signed payload (HMAC-SHA256)
IMEI Registry Sync	IMEI status updates: blacklisted, whitelisted, under-investigation	RURA Registry ↔ iNgabo	REST API (pull) + webhook (push) + mTLS	mTLS + API key + RURA PKI certificate
SMS Metadata Stream	SMS event metadata: sender hash, recipient hash, message length, timestamp, bulk-send flag	MNO → iNgabo (SMSC tap)	Kafka Connect over TLS	mTLS + hash pseudonymization at SMSC
Voice CDR Metadata	Call metadata only — no audio: duration, CLI, destination, cell ID, roaming, call type	MNO → iNgabo	Kafka Connect over TLS	mTLS; strict prohibition on content data

Integration Layer	Data Exchanged	Direction	Protocol	Security
CAMARA API (outbound)	Network capability queries as defined in §8	iNgabo → MNO CAMARA endpoint	REST HTTPS + OAuth2 Bearer + mTLS	mTLS + purpose-code-bound tokens
Fraud Alert Dispatch (outbound)	Fraud alerts to MNO security teams: alert type, MSISDN hash, severity, recommended action	iNgabo → MNO	REST webhook + mTLS	mTLS + signed payload

9.2 CDR Ingestion Pipeline



Figure 11: CDR Ingestion Pipeline - From MNO to Fraud Detection

9.3 SS7 and DIAMETER Intelligence

iNgabo's telecom protocol intelligence is currently scoped to CDR metadata and CAMARA API integration. Direct SS7/DIAMETER signaling layer monitoring, while technically planned for Phase 3, requires formal RURA authorization and MNO signaling infrastructure access agreements beyond CDR data sharing. The following describes the Phase 3 planned capability:

REGULATORY NOTE - SS7/DIAMETER Access

Direct SS7/DIAMETER signaling plane access requires RURA authorization under the Electronic Communications Act and formal technical agreements with each MNO. This capability is not deployed in Phase 1 or Phase 2. The architecture is designed to accommodate this integration when regulatory authorization is obtained in Phase 3.

Table 21: Signaling Fraud Detection: Phase 2 vs Phase 3 Capability

Signaling Attack Pattern	Detection Method (Phase 2 — CDR-based)	Detection Method (Phase 3 — Signaling)	Severity
Location update storms (SS7 MAP)	CDR: abnormal cell-tower registration frequency	SS7: MAP UpdateLocation request volume anomaly	High
Subscriber redirection / hijack	CDR: traffic pattern change post-swap	DIAMETER: MME registration anomaly	Critical
IMSI catching / fake base station	CDR: abnormal cell ID appearance, signal strength anomaly (via MNO)	SS7: IMSI detach/attach anomaly	Critical

Signaling Attack Pattern	Detection Method (Phase 2 — CDR-based)	Detection Method (Phase 3 — Signaling)	Severity
Wangiri / one-ring fraud	CDR: single-ring calls to premium international destinations	Same - CDR-sufficient for detection	High
CLI spoofing / number impersonation	CAMARA: Number Verification cross-check	SIP/ISUP: CLI vs. network-asserted identity mismatch	High
DIAMETER EAP fraud	Not detectable via CDR	DIAMETER: EAP-AKA/SIM auth anomaly detection	High

9.4 MNO Onboarding Process

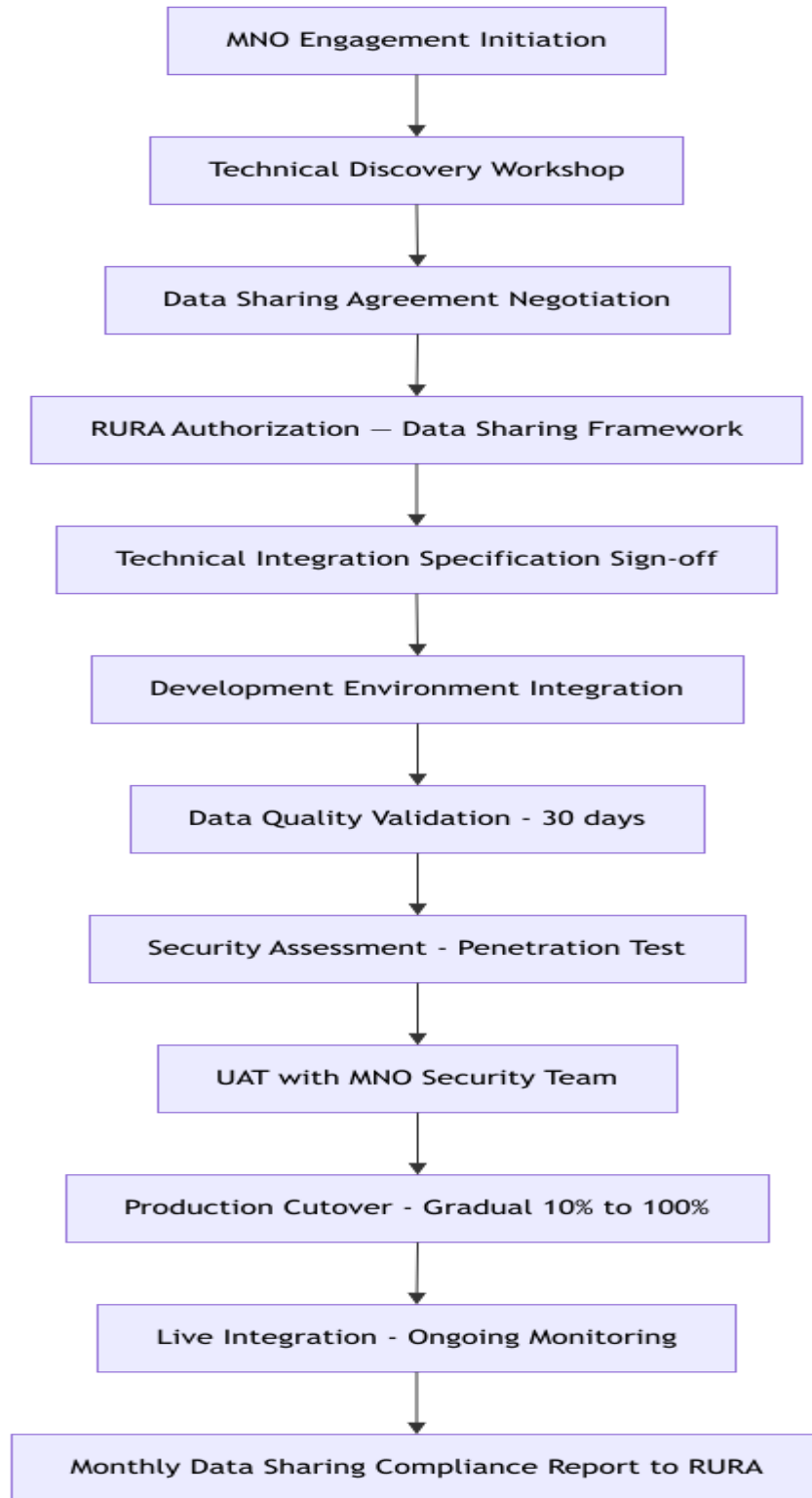


Figure 12: MNO Integration

Onboarding Process

10. PHONE THEFT RECOVERY ARCHITECTURE



10.1 Architecture Overview

The phone theft recovery system integrates IMEI intelligence, network-assisted device tracking (within legal frameworks), SIM replacement detection, and investigation workflow management into a unified platform accessible by RIB investigators and Rwanda National Police dispatch teams.

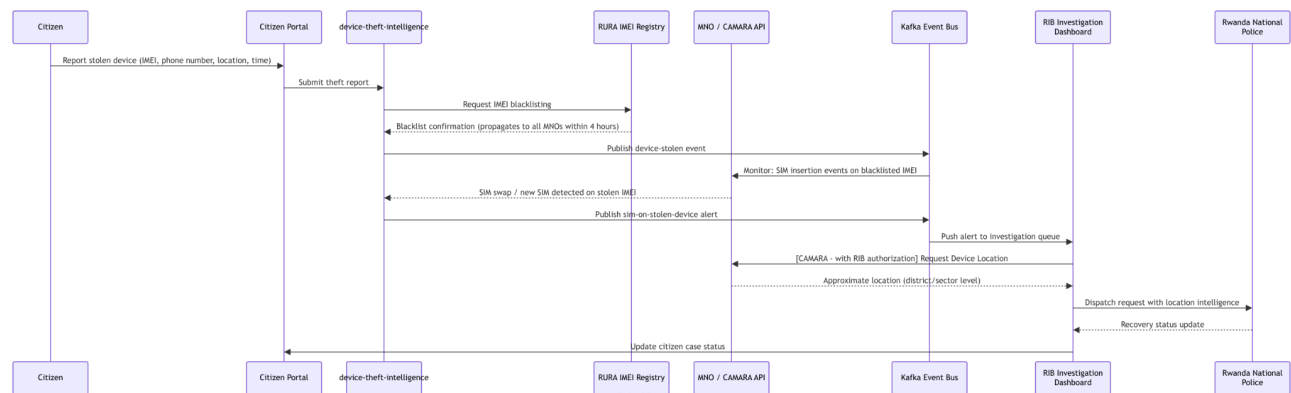


Figure 13: Phone Theft Recovery - End-to-End Workflow Sequence

10.2 Theft Report Data Model

```
// Theft Report Entity Schema
{
  "report_id": "RPT-2026-0001234", // Platform-generated case reference
  "reported_at": "2026-06-15T14:23:11Z",
  "status": "ACTIVE", // ACTIVE | RECOVERED | CLOSED | UNRESOLVED
  "device": {
    "imei": "[HMAC-SHA256 hash - stored only]",
  }
}
```

```

    "imei_display": "35812309XXXXXX",           // Partially masked for display
    "make": "Samsung",
    "model": "Galaxy A23",
    "color": "Black",
    "os": "Android 13"
  },
  "owner": {
    "citizen_id": "[encrypted foreign key to citizen record]",
    "msisdn_hash": "[HMAC-SHA256 hash]",
    "national_id_hash": "[HMAC-SHA256 hash]"
  },
  "incident": {
    "theft_datetime": "2026-06-15T12:00:00Z",
    "location": { "district": "Kicukiro", "sector": "Niboye", "lat_approx": -1.98,
    "lon_approx": 30.10 },
    "police_station": "Kicukiro Police Station",
    "police_ref": "KIC/2026/001234"
  },
  "blacklist_status": {
    "requested_at": "2026-06-15T14:25:00Z",
    "rura_confirmation": "RURA-BL-2026-054321",
    "mno_propagation": [
      { "mno": "MTN Rwanda", "confirmed_at": "2026-06-15T15:10:00Z" },
      { "mno": "Airtel Rwanda", "confirmed_at": "2026-06-15T16:02:00Z" }
    ]
  },
  "investigation": {
    "rib_case_id": "RIB-CYB-2026-0045",
    "assigned_investigator": "[officer_id]",
    "priority": "MEDIUM",
    "events": [] // Chronological investigation event log
  }
}

```

Figure 14: Theft Report Entity Schema - Pseudonymized Data Model

10.3 IMEI Intelligence Engine

Table 22: IMEI Intelligence Capabilities and Legal Framework

Capability	Implementation	Data Source	Legal Framework
IMEI blacklisting	API call to RURA national IMEI registry; confirmation webhook back to platform	RURA IMEI Registry API	RURA Electronic Communications Act; theft report as trigger
Blacklist propagation monitoring	Track MNO acknowledgment of RURA blacklist update via webhook	MNO APIs	Included in MNO Data Sharing Agreement

Capability	Implementation	Data Source	Legal Framework
SIM-on-stolen-IMEI detection	CDR stream: detect CDR activity on blacklisted IMEI hash	MNO CDR stream	MNO Data Sharing Agreement; RURA authorization
New SIM insertion detection	CAMARA Device Status API: check registration status changes on known IMEI	CAMARA API (Phase 2)	RURA Data Sharing Agreement + RIB case reference
IMEI cloning detection	Statistical: detect multiple network presences of same IMEI simultaneously	MNO CDR streams (multi-operator)	MNO Data Sharing Agreements + RURA authorization
Network location (approximate)	CAMARA Device Location API: district/sector granularity	CAMARA API (Phase 2/3)	RIB case reference required per API call; RURA authorization
Network location (precise)	CAMARA Device Location API: precise coordinates (Phase 3)	CAMARA API (Phase 3)	Court order or equivalent RIB legal instrument required
Cross-border tracking flag	Detect if stolen device SIM becomes active on roaming network in EAC	CDR roaming records + EAC intelligence exchange (Phase 5)	Bilateral EAC digital cooperation agreement

10.4 Chain of Custody Management

All evidence collected through the iNgabo platform - device location records, CDR metadata, SIM event logs, CAMARA API responses - is stored in a chain-of-custody compliant evidence store with immutable audit logging, cryptographic integrity verification, and export capability meeting Rwandan court admissibility standards.

Table 23: Chain of Custody Evidence Management

Evidence Type	Storage	Integrity Mechanism	Access Control	Export Format
CAMARA API response records	ClickHouse (immutable, write-once table)	SHA-256 hash of each record; daily Merkle tree root signed by platform HSM	RIB investigators + legal team; logged access	PDF/A for court; JSON for digital systems
CDR metadata records	ClickHouse (immutable)	Same as above	Senior investigator + supervisor approval	Structured report PDF; CSV with integrity hash
SIM event logs	PostgreSQL (append-only with trigger)	Row-level HMAC; deletion blocked at DB trigger level	RIB case lead only	Court report PDF with chain-of-custody attestation
Investigation actions log	ClickHouse (write-once)	Kafka audit events → ClickHouse; no update/delete API	Audit team + senior management only	Audit export PDF with digital signature

Evidence Type	Storage	Integrity Mechanism	Access Control	Export Format
Network location records	PostgreSQL (encrypted, append-only)	HSM-signed location records with timestamp and requesting officer ID	RIB case lead + supervisor; court order documented	Court-admissible location report with legal attestation

11. SMS INTELLIGENCE ENGINE

SECTION 11

SMS INTELLIGENCE ENGINE

Smishing Detection, Phishing Analysis, and Campaign Intelligence



11.1 Architecture Overview

The SMS Intelligence Engine processes SMS event metadata from MNO SMSC streams, applying NLP-based content classification, URL threat analysis, sender reputation scoring, and campaign fingerprinting to detect smishing, phishing, bank impersonation, government impersonation, and bulk scam campaigns in real time.

PRIVACY DESIGN NOTE

SMS content analysis requires MNO authorization and is subject to subscriber privacy protections under Rwanda Law 058/2021. Content analysis is performed for identified threat categories only; general SMS content is not stored. All message content processed for classification is immediately discarded after classification - only the classification result, metadata, and extracted threat indicators are retained.

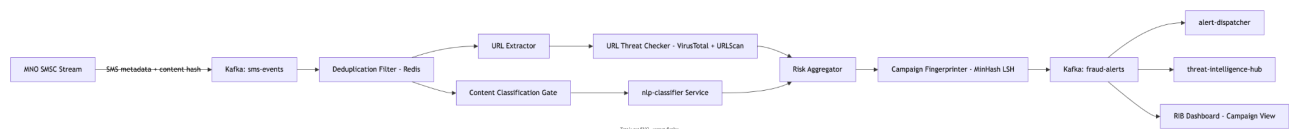


Figure 15: SMS Intelligence Engine Processing Pipeline

11.2 Classification Schema

Table 24: SMS Classification Schema and Response Actions

Class	Sub-classes	Action	Citizen Notification
SAFE	-	No action; metadata retained for behavioral baseline	None
SUSPICIOUS	Unusual sender, possible promotional spam, borderline content	Enhanced monitoring; sender reputation downgrade	Optional - citizen can enable suspicious SMS alerts

Class	Sub-classes	Action	Citizen Notification
FRAUDULENT	Bank impersonation, government impersonation, lottery scam, advance fee, phishing link	Block (if MNO has blocking API); MNO alert; public campaign database update	SMS alert to targeted subscriber if identifiable
CRITICAL	Active campaign with confirmed victims, known criminal infrastructure URL, national institution impersonation	Immediate MNO escalation; RIB notification; public warning issued; NCSA notification	Emergency alert to all subscribers who received this campaign

11.3 Campaign Detection

The campaign fingerprinter uses MinHash Locality Sensitive Hashing (LSH) to detect near-duplicate SMS messages sent at volume - the signature of an organized smishing campaign. Unlike exact-match deduplication, MinHash LSH identifies messages that are structurally similar but with minor variations (different URLs, different amounts, different victim names) that evade keyword filters.

Table 25: SMS Campaign Detection Techniques

Detection Technique	Target	Threshold	Time Window
MinHash LSH near-duplicate detection	Bulk smishing campaigns with template variations	> 50 near-duplicate messages from common sender set	1 hour sliding window
Sender reputation clustering	SIM farm / bulk SMS sender networks	> 20 messages per MSISDN per hour to different recipients	1 hour
URL repeat detection	Phishing URL appearing in multiple independent messages	Same URL hash appearing > 5 times from different senders	24 hours
Impersonation entity detection	Messages claiming to be from BNR, RURA, MTN, Airtel, Government	NLP entity extraction: organization claim vs. sender verification	Real-time
Cross-operator campaign correlation	Campaigns spanning multiple MNO networks	Same fingerprint cluster detected on 2+ operator streams	12 hours

12. VOICE CALL INTELLIGENCE ENGINE

SECTION 12

VOICE INTELLIGENCE ENGINE

Vishing Detection, Caller Reputation, and Metadata Analytics



12.1 Scope and Legal Framework

IMPORTANT - LEGAL SCOPE DEFINITION

The iNgabo Voice Intelligence Engine operates exclusively on voice call METADATA. No voice call audio content is recorded, processed, analyzed, or stored by this platform without explicit legal authorization from a competent Rwandan court. Voice metadata analysis (call duration, frequency, origination pattern, CLI analysis) does not constitute interception under the applicable legal framework. Any future capability involving voice content analysis will require separate legal authorization, RURA approval, and NCSA security clearance before deployment.

12.2 Voice Metadata Intelligence

The voice intelligence engine analyzes patterns in voice call metadata to detect vishing campaigns, social engineering operations, and organized fraud call centers - without accessing call content. The following signals are derived exclusively from CDR metadata:

Table 26: Voice Metadata Intelligence Signals

Signal	Derived From	Fraud Pattern Detected	ML Feature
Call duration distribution	CDR: call duration field	Vishing calls typically 2–8 minutes (script duration); robocalls < 30 seconds	Duration histogram vs. baseline
Sequential dialing pattern	CDR: call sequence, timing	Autodialers call numbers sequentially; victim targeting shows clustering	Inter-call interval distribution
One-ring pattern (Wangiri)	CDR: call duration < 5 seconds, international origin	Wangiri callback scam: victim calls expensive premium number back	Ring-and-drop event count per number

Signal	Derived From	Fraud Pattern Detected	ML Feature
Geographic origin clustering	CDR: international CLI, roaming origin	Cross-border fraud call centers showing calls from specific countries	Origination country concentration index
Callback rate anomaly	CDR: call pairs (outbound followed by inbound)	Wangiri success rate: how many victims called back	Callback rate per origination number
CLI mismatch	CDR: CLI vs. CAMARA Number Verification response	Spoofed numbers impersonating legitimate institutions	CLI vs. network-asserted identity delta
Call velocity to unique victims	CDR: unique destination count per origination MSISDN per hour	High-volume fraud call campaigns	Destination diversity per hour
Victim repeat contact	CDR: same destination contacted multiple times in short window	Social engineering persistence — repeated contact of same victim	Repeat contact frequency

12.3 Vishing Campaign Detection

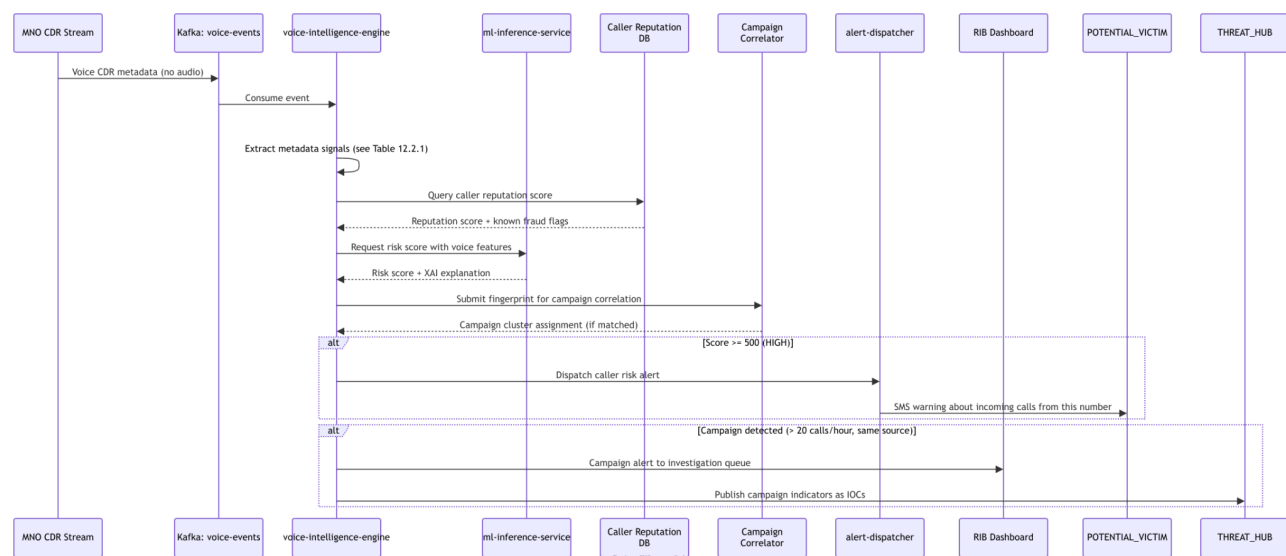


Figure 16: Vishing Campaign Detection - Sequence Diagram

12.4 Caller Reputation System

The caller reputation system maintains a dynamic reputation score for every MSISDN that has been flagged in the iNgabo platform. Reputation scores decay over time (no permanent blacklisting without human review) and are updated in real-time as new fraud signals arrive.

Table 27: Caller Reputation Score Tiers

Score Range	Label	Citizen Warning Shown	LEA Visibility
0 – 199	CLEAN	None	Not visible
200 – 399	WATCH	Optional caller ID: "Unknown caller - no fraud history"	Accessible to analysts with justification
400 – 599	SUSPICIOUS	Caller ID warning: "This number has suspicious activity"	Visible in analyst workbench
600 – 799	PROBABLE FRAUD	Strong warning: "High fraud risk caller - do not share personal information"	Highlighted in investigation queue
800 – 1000	KNOWN FRAUD	Alert: "Known fraud number - this call is likely a scam"	Immediate RIB notification; case auto-created

13. FRAUD INTELLIGENCE ENGINE

SECTION 13

FRAUD INTELLIGENCE ENGINE

Multi-Signal Correlation, Threat Intelligence, and Pattern Library



13.1 Engine Architecture

The Fraud Intelligence Engine (FDE) is the central orchestrator of iNgabo's fraud detection capability. It consumes events from all intelligence feeds - CDR, SMS, voice, device, transaction - correlates signals across sources, applies a layered detection stack (rules → ML → graph), and produces a unified fraud risk assessment with XAI explanation and recommended action.

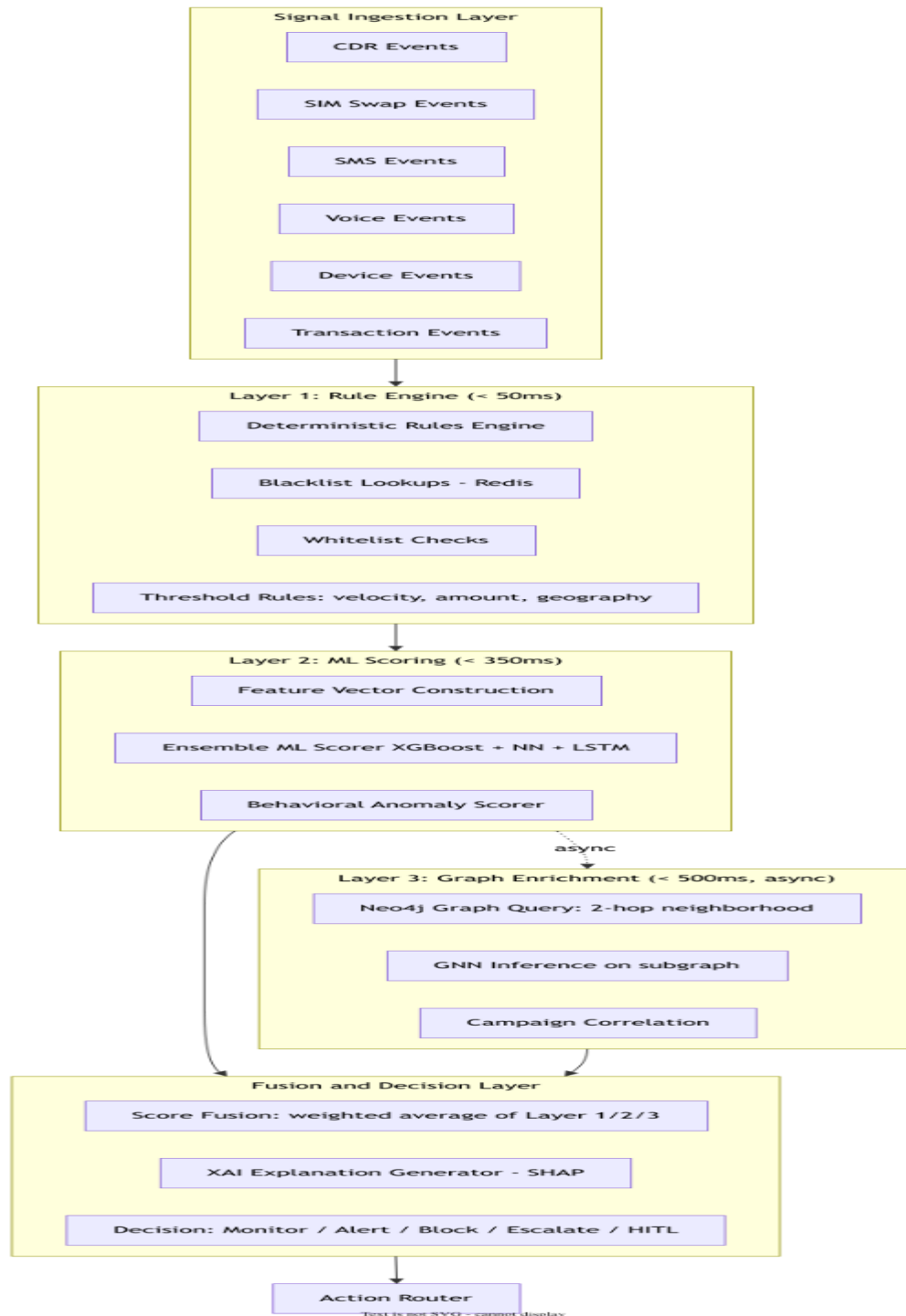


Figure 17: Fraud Detection Three-Layer Pipeline Architecture

13.2 Rule Engine

The rule engine applies deterministic, interpretable rules that operate on every event within a 50ms budget. Rules are stored in a versioned rule repository (git-backed) and loaded into memory at startup with live reload

capability. Rules are categorized by type and can be enabled/disabled per deployment configuration without code deployment.

Table 28: Rule Engine Categories and Actions

Rule Category	Example Rules	Action on Match	Tunable
Blacklist	IMEI in RURA blacklist; MSISDN in fraud blacklist; URL in threat feed	Immediate flag — escalate to Layer 2 with pre-score 600+	No - regulatory
Velocity	> 20 unique call destinations in 1 hour; > 10 transactions in 15 minutes	Score boost + enhanced monitoring	Yes - per deployment
Geography	Cell tower transition distance > 500km in < 30 minutes	Score boost - geographic impossibility	Yes - threshold
SIM Age	SIM age < 7 days AND high-value transaction > RWF 500,000	Score boost - new SIM + high value	Yes - thresholds
Device Pairing	IMEI used with > 3 different SIMs in 30 days	Score boost - SIM farm indicator	Yes - threshold
Time Anomaly	Transaction at 01:00–04:00 AND > 2 standard deviations from baseline	Score boost - unusual time activity	Yes - hours range
Known Scam Pattern	SMS contains keyword cluster matching known scam templates	Immediate FRAUDULENT classification	Yes - keyword library
International CLI	Call from international CLI matching known vishing country patterns	Score boost - caller origin risk	Yes - country list

13.3 Fraud Pattern Library

The fraud pattern library maintains signatures for known fraud attack patterns, campaign templates, and criminal tradecraft. Patterns are updated from four sources: GSMA T-ISAC feeds, RIB intelligence submissions, NCSA threat reports, and platform-discovered patterns confirmed by analyst review.

Table 29: Fraud Pattern Library - Phase 4 Targets

Pattern Category	Count (Phase 4 Target)	Update Source	Confidence Rating
SMS Scam Templates (Kinyarwanda)	500+	Platform discovery + RIB submissions	High — native language validation
SMS Scam Templates (English/French)	1,200+	GSMA T-ISAC + platform discovery	High
Phishing URL Patterns	5,000+ domain patterns	URLScan.io + VirusTotal + OpenPhish	Very High — external validation

Pattern Category	Count (Phase 4 Target)	Update Source	Confidence Rating
Known Fraud MSISDN (active)	50,000+	RIB submissions + platform detection + MNO reports	High
IMEI Fraud Registry (cloned/modified)	25,000+	RURA registry + platform detection	Very High — RURA authoritative
Vishing Number Patterns	10,000+	CDR analysis + citizen reports + RIB	Medium — behaviorally derived
Fraud Campaign Fingerprints	2,000+	MinHash corpus from campaign detection	High — platform-generated
Transaction Fraud Sequences	300+ sequence types	Graph analysis + supervised labeling	High

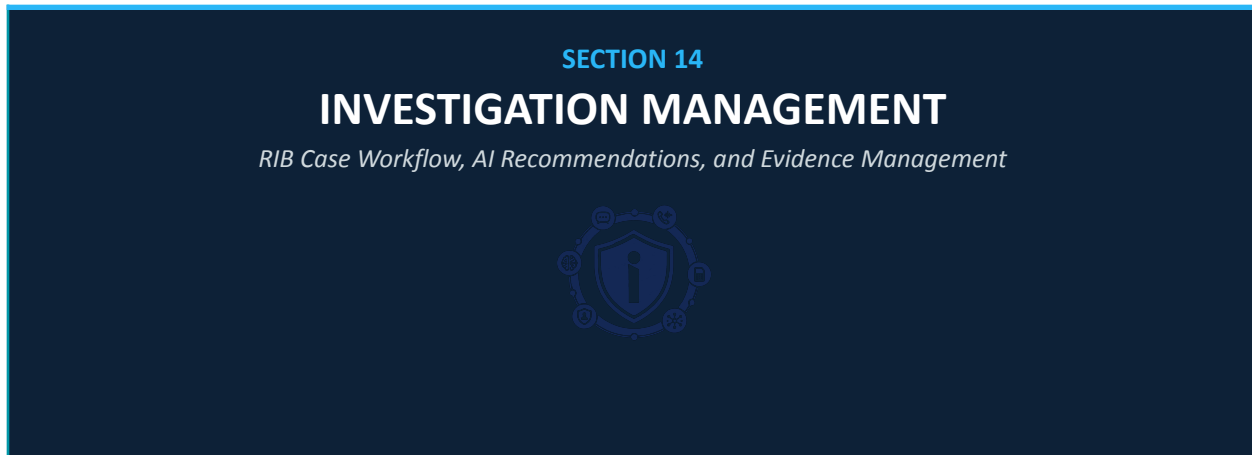
13.4 Cross-Signal Correlation

The most powerful detections emerge when signals from multiple channels are correlated. The FDE maintains a real-time correlation window that links events across channels for the same entity within a configurable time window (default: 24 hours):

```
// Cross-signal correlation example: SIM Swap + MoMo transaction + SMS campaign
//
// Signal 1 (T+0): SIM swap on MSISDN A → risk score 720 (HIGH)
// Signal 2 (T+12min): Mobile money transaction from MSISDN A to unfamiliar account
//                      Amount: RWF 850,000 (> 3 std dev above baseline)
// Signal 3 (T+15min): SMS from MSISDN B to MSISDN A matching bank impersonation
//                      template
//
// Correlation engine detects: SIM swap BEFORE high-value transaction AFTER
// impersonation SMS
// → This is the canonical account-takeover-via-SIM-swap sequence
// → Correlated score: 940 (IMMINENT) — exceeds individual signal scores
// → Actions: Block MoMo transaction (if MoMo integration active)
//             Alert citizen on old SIM
//             Notify RIB
//             Auto-create investigation case
//             HITL queue with senior analyst notification
```

Figure 18: Cross-Signal Correlation - SIM Swap Account Takeover Sequence Example

14. INVESTIGATION MANAGEMENT SYSTEM



14.1 Investigation Platform Architecture

The Investigation Management System (IMS) provides Rwanda Investigation Bureau and Rwanda National Police cybercrime units with a structured digital case management environment backed by iNgabo's fraud intelligence graph. The IMS is not a generic case management tool - it is purpose-built for telecom fraud and cybercrime investigation, with deep integration to the fraud intelligence, graph, and AI recommendation layers.

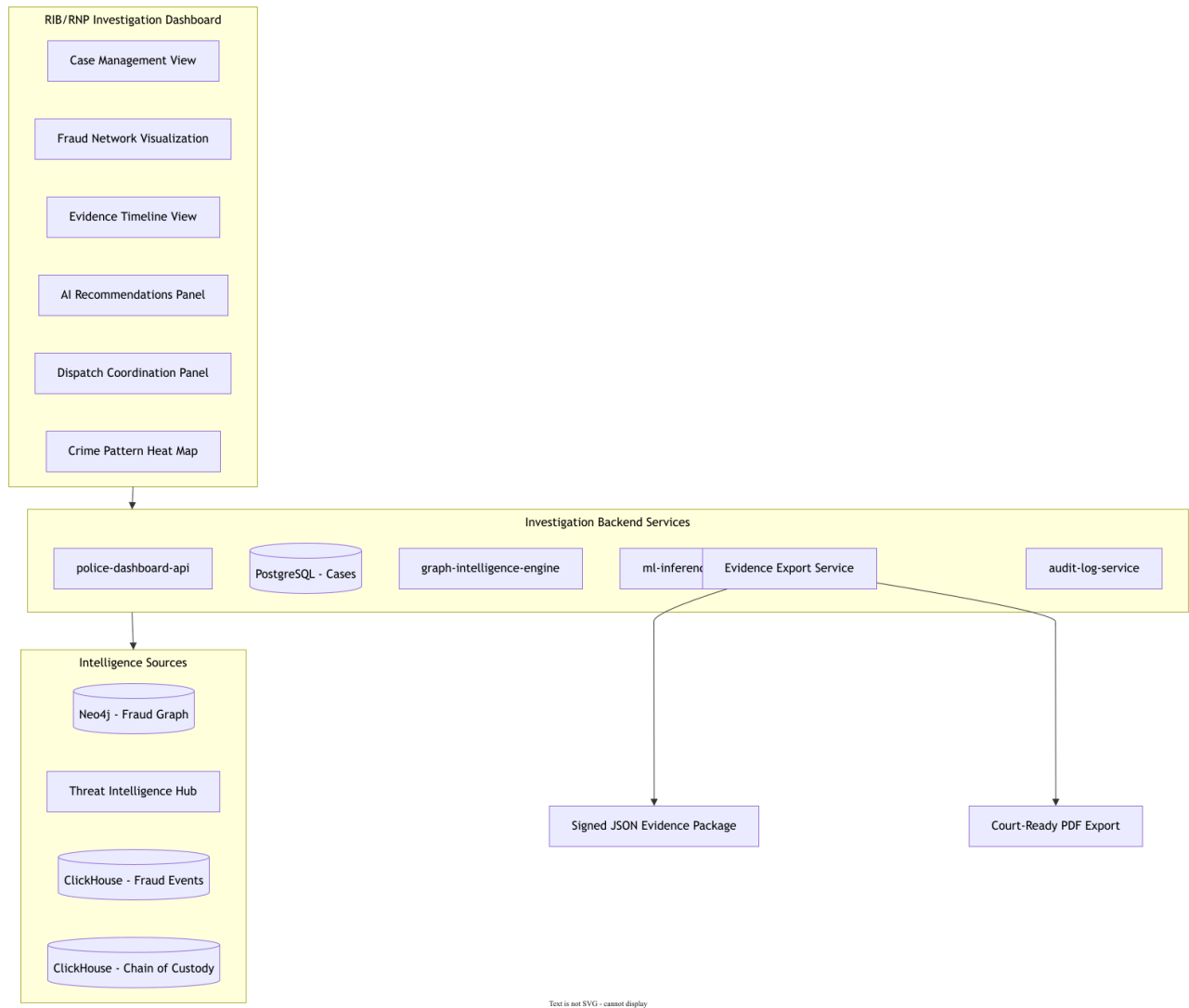


Figure 19: Investigation Management System Architecture

14.2 Case Lifecycle

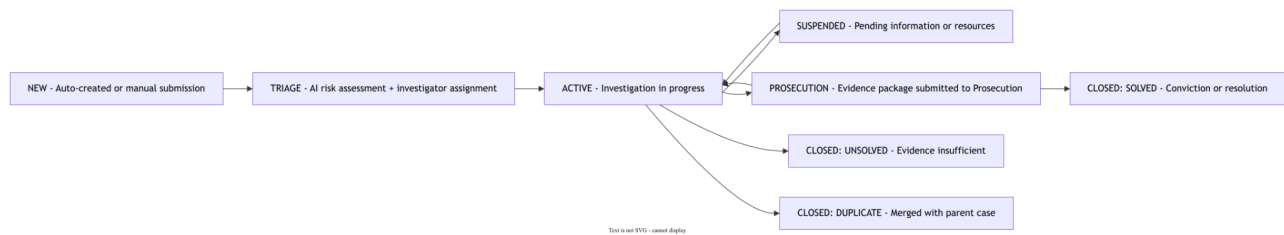


Figure 20: Investigation Case Lifecycle State Machine

14.3 AI Investigation Recommendations

The AI recommendation engine analyzes active cases and generates structured investigation recommendations based on pattern matching against resolved cases, graph analysis of connected entities, and ML-based prediction of investigation effectiveness.

Table 30: AI Investigation Recommendation Types

Recommendation Type	Trigger	AI Basis	Human Decision Required
Entity expansion	New case created	Graph: 2-hop neighborhood contains previously flagged entities → "Investigate connected MSISDN X"	Investigator must approve before adding to case scope
Case linkage	Similar pattern detected in another case	ML: case feature similarity > 0.85 → "This case matches RIB-CYB-2026-0023 — consider merging"	Senior analyst approval for case merge
Priority escalation	High-value victim or critical infrastructure involved	Rule: victim is government official / financial institution → "Escalate to senior investigator"	Supervisor acknowledgment
Interview recommendation	Suspect identified in graph with high centrality	Graph: PageRank top 5 in fraud cluster → "Priority interview: MSISDN +250 7XX XXX XXX"	Investigator judgment
Evidence preservation	Key evidence approaching retention limit	Rule: evidence retention < 30 days → "Urgent: preserve evidence before retention expiry"	Investigator action required
Cross-border flag	Evidence of EAC-region involvement	CDR: roaming activity + international CLI → "Consider Interpol I-24/7 notification"	Legal review + senior approval

14.4 Evidence Export and Court Packaging

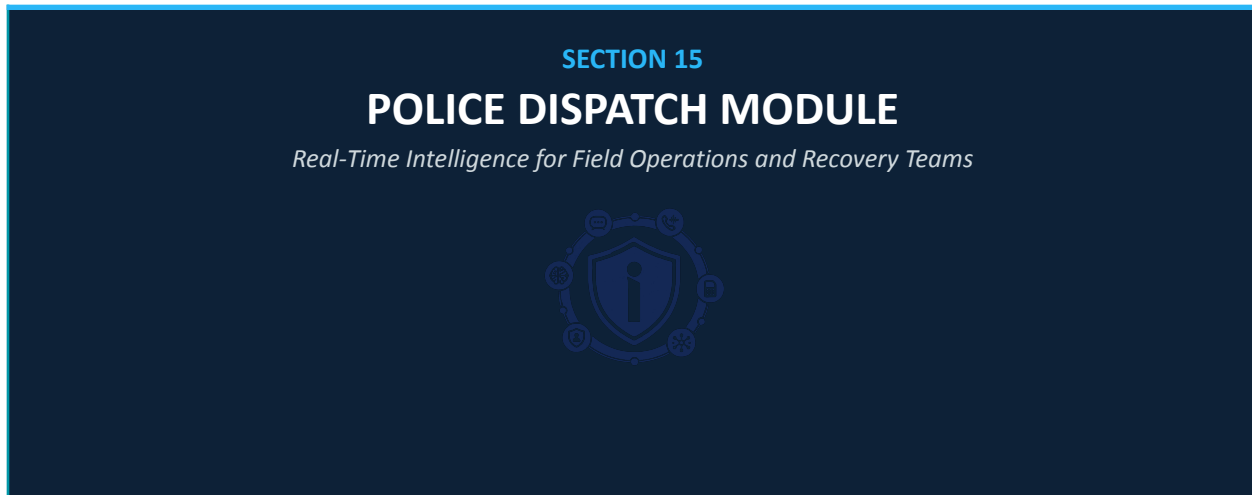
The Evidence Export Service generates court-ready evidence packages that meet Rwandan court admissibility standards. Each package includes a digital integrity attestation signed by the platform HSM, a complete chain-of-custody log, and investigator declaration sections for signature.

Table 31: Court Evidence Package Components

Package Component	Format	Integrity Control	Purpose
Case Summary Report	PDF/A (archival)	Platform HSM digital signature + SHA-256 manifest	Primary court submission document
Evidence Timeline	PDF/A with embedded JSON	Record-level HMAC verification	Chronological evidence presentation

Package Component	Format	Integrity Control	Purpose
Network Relationship Graph	PDF visual + Cypher export	Graph snapshot hash + timestamp	Criminal network structure for court
CDR Metadata Records	CSV + integrity manifest	SHA-256 per record + Merkle root	Supporting telecom evidence
API Response Records (CAMARA)	JSON + PDF attestation	HSM-signed response records	Network intelligence evidence
Audit Log Extract	PDF + raw ClickHouse export	ClickHouse system digest + row hashes	Access and action chain of custody
Investigator Declaration	Word template (for signature)	N/A — requires wet/digital signature	Legal accountability declaration

15. POLICE DISPATCH MODULE



15.1 Dispatch Module Design

The Police Dispatch Module provides Rwanda National Police field commanders and dispatch coordinators with real-time operational intelligence for device recovery operations and fraud incident responses. The module emphasizes speed, map-based interface, and minimal information complexity - designed for operational use by officers in the field.

Table 32: Police Dispatch Module Features

Dispatch Feature	Description	Data Source	Update Frequency
Live incident map	Heat map of active fraud alerts, stolen device locations, and ongoing crime patterns by district/sector	Fraud alert stream + device location (CAMARA, authorized)	Real-time, 30-second refresh
Active case queue	Ranked list of cases requiring field action, sorted by priority and proximity to available units	Investigation management system case database	Real-time push from case management
Device location intel	Approximate district/sector location of targeted device (within legal authorization)+live gps coordinates->best	CAMARA Device Location API (RIB-authorized)	On-demand per case reference
Unit status board	Real-time status of deployed field units and available capacity	Field unit app (mobile)	Real-time push from officer devices
Recovery task assignment	Assign recovery task to nearest available unit with case brief and location intel	Case management + unit status	Real-time — < 5 second assignment notification

Dispatch Feature	Description	Data Source	Update Frequency
Evidence checklist	Field evidence collection checklist calibrated for device recovery operations	Case template engine	Per-case, generated at task assignment
Communication log	Secure messaging between dispatch, field units, and RIB case officer	Encrypted in-platform messaging	Real-time
Outcome recording	Field officer records recovery outcome; updates case status in real time	Mobile app → police-dashboard-api	Real-time sync on field action

15.2 Dispatch Workflow

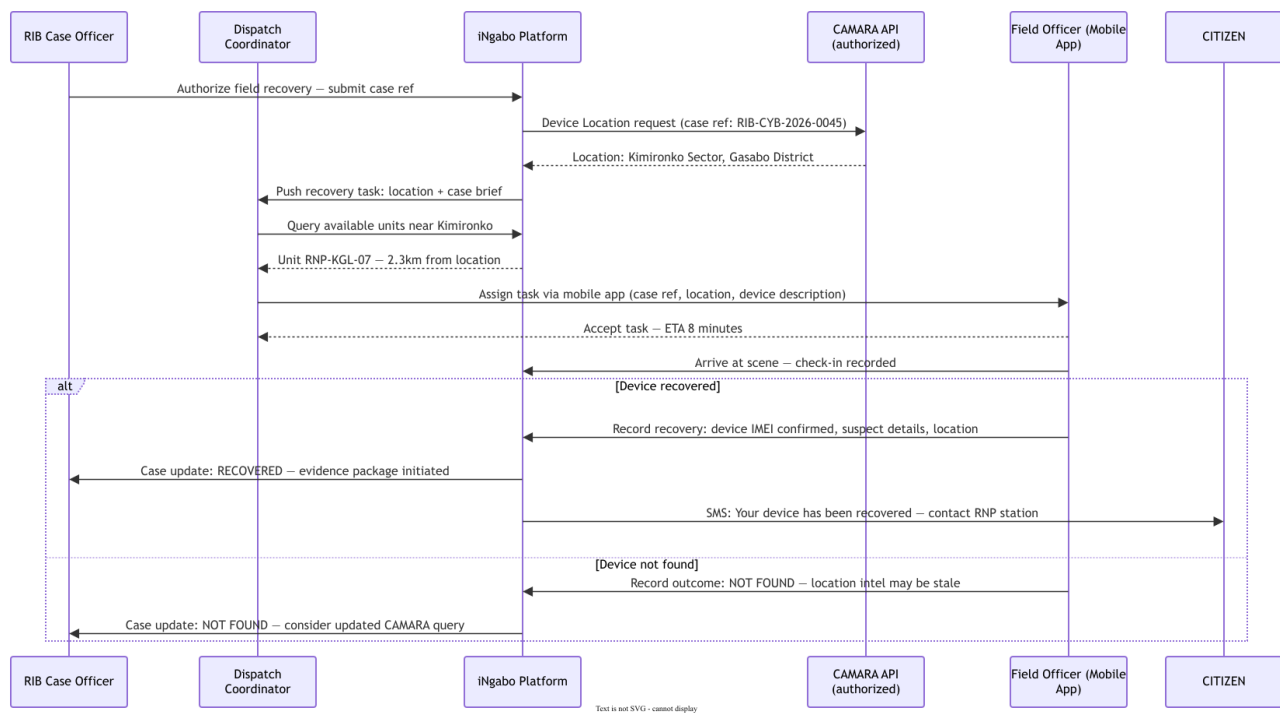


Figure 21: Police Dispatch - Field Recovery Operation Sequence

16. CITIZEN SECURITY PLATFORM



16.1 Multi-Channel Architecture

The Citizen Security Platform is the public face of iNgabo - the direct protection layer for every Rwandan. Recognizing that digital access varies dramatically across Rwanda, the platform delivers critical protection functions across four channels: web portal, mobile app (iOS and Android), USSD (*900# via feature phone), and proactive SMS alerts with no citizen action required.

Table 33: Citizen Security Platform Channel Architecture

Channel	Technology	Target Users	Key Functions	Phase
Web Portal	React.js, Progressive Web App	Smartphone + computer users	Full feature set; fraud reporting; case tracking; IMEI check; number verification; education content	Phase 1
Mobile App	React Native (iOS + Android)	Smartphone users	Fraud alerts; number scan; case tracking; USSD bridge; push notifications	Phase 2
USSD (*900#)	Africa's Talking USSD API, NestJS	Feature phone users - all subscribers	Number verification; theft report; scam report; case status check	Phase 2
Proactive SMS Alerts	Africa's Talking SMS API	All subscribers (no registration required)	Fraud warnings; SIM swap alerts; campaign warnings; recovery updates	Phase 2

16.2 USSD Interface Design

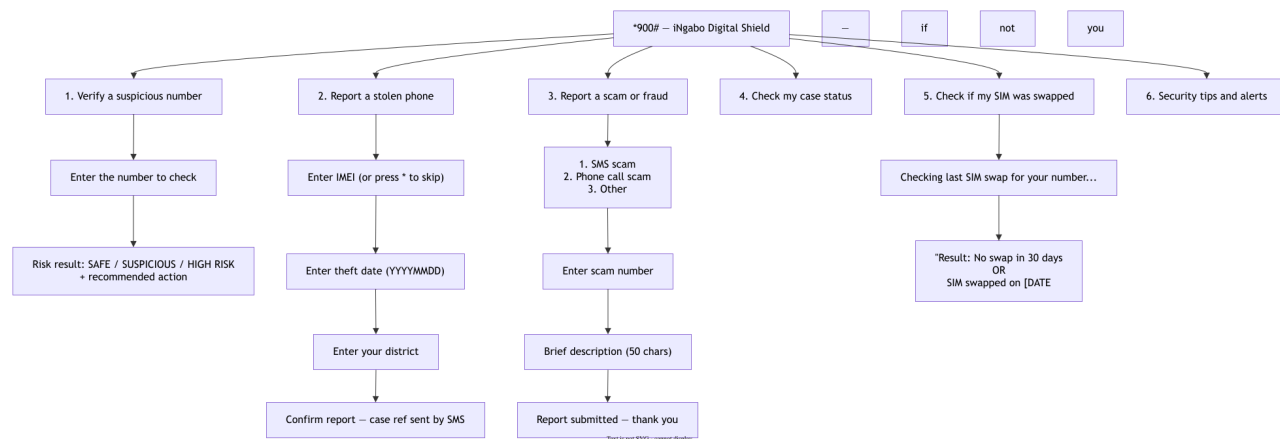


Figure 22: USSD Menu Architecture — *900# Citizen Interface

16.3 Citizen Data Model and Privacy

Citizens interact with iNgabo at three authentication levels. Authentication level determines data access and personalization - unauthenticated users can still access core protection functions (number verification, scam reporting) without creating an account.

Table 34: Citizen Authentication Levels

Auth Level	Registration Req.	Available Functions	Data Stored
Anonymous	None	Number verification, scam report (no tracking), public scam database	No personal data — IP pseudonymized; report stored without citizen link
USSD / SMS Verified	MSISDN verification only	All USSD functions, SMS alerts, theft report, case tracking by MSISDN	MSISDN hash, case references — no name or identity data
Full Account (Irembo-linked)	National ID via Irembo integration	Full web/app access, detailed case history, personalized alerts, enhanced SIM swap monitoring	Encrypted national ID hash, MSISDN hash, case history, alert preferences

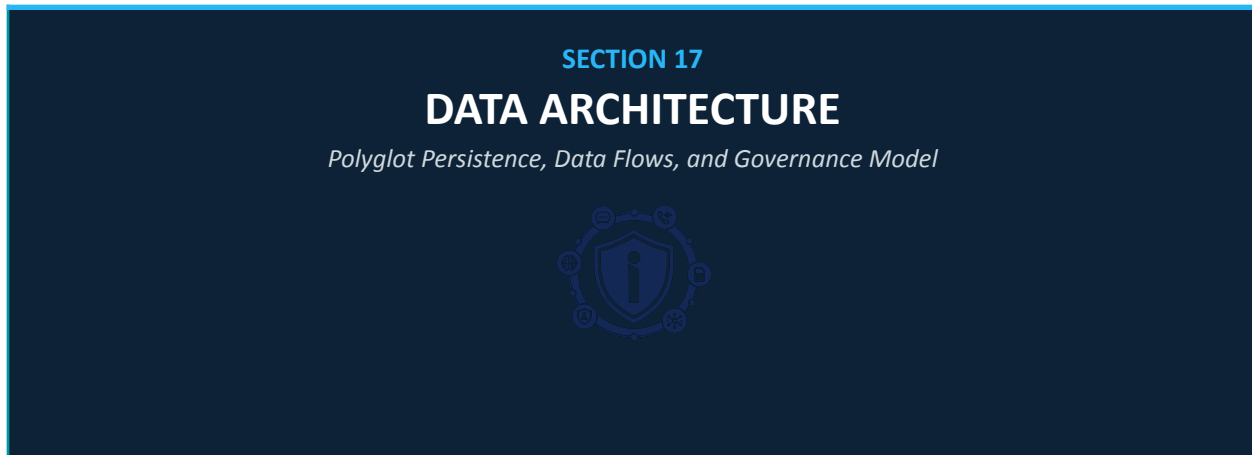
16.4 Proactive Alert Design

Proactive alerts are the highest-impact citizen protection function - they protect citizens who have not registered with the platform and who are not actively checking for threats. The alert system monitors all fraud intelligence streams and dispatches targeted warnings to at-risk MSISDNs as soon as a threat is detected.

Table 35: Citizen Proactive Alert Types

Alert Type	Trigger	Delivery Channel	Message Example
SIM Swap Alert	SIM swap request risk score ≥ 500 on citizen's MSISDN	SMS to old SIM immediately	iNgabo Alert: A SIM swap was attempted on your number. If this was NOT you, call 3909 immediately or dial *900#.
Fraud Caller Warning	Incoming call from MSISDN with reputation score ≥ 600	SMS before/during call (via CDR if technically feasible with MNO)	iNgabo: The number just calling you (+250 78X XXX) is flagged for fraud. Do not share personal or account information.
Smishing Campaign Warning	Citizen's MSISDN is in the recipient list of a detected smishing campaign	SMS immediately on campaign detection	iNgabo Alert: A scam SMS may be sent to you soon from [number]. It is NOT from [impersonated organization]. Delete it and report to *900#.
Stolen Device Activity	New SIM detected on citizen's reported stolen device	SMS to citizen's alternate number or email	iNgabo: Your stolen phone (Samsung Galaxy A23) is active on the network in Kicukiro District. RNP has been notified. Case ref: RPT-2026-001234.
Account Takeover Warning	Correlated SIM swap + high-value MoMo transaction pattern	Emergency SMS to citizen	URGENT iNgabo Alert: Suspicious account activity after recent SIM event. Check your accounts immediately. Call 3909 if you did not authorize this activity.

17. DATA ARCHITECTURE



17.1 Polyglot Persistence Strategy

iNgabo employs a purpose-fit polyglot persistence strategy: each data domain uses the storage technology best matched to its access patterns, consistency requirements, and query characteristics. No single database technology serves all needs; the data layer is a deliberate ensemble of complementary stores.

Table 36: Polyglot Persistence — Data Store Selection

Data Store	Technology	Version	Primary Use	Data Volume (Phase 4)
Transactional / Operational	PostgreSQL	15+ with TimescaleDB ext.	Cases, citizens, reports, configurations, operational state	~500GB — moderate growth
Fraud Intelligence Graph	Neo4j	5.x Enterprise (Causal Cluster)	Entity relationships, fraud networks, investigation graphs	~2TB — high relationship density
Real-time Cache	Redis	7.x Cluster Mode	Risk score cache, blacklist hot copy, session data, rate limiting	~50GB — TTL-bounded
Event Streaming	Apache Kafka	3.x (KRaft mode)	All platform events, audit events, inter-service communication	Retained 90 days on disk — ~5TB
Fraud Event Analytics	ClickHouse	23.x Cluster	Fraud event history, audit log, behavioral analytics, regulatory reports	~10TB — 7-year retention
Behavioral Time Series	TimescaleDB (PostgreSQL ext.)	2.x	Per-entity behavioral baselines, time-series anomaly detection	~800GB

Data Store	Technology	Version	Primary Use	Data Volume (Phase 4)
ML Model Store	MLflow + S3-compatible (MinIO)	Latest	Trained model artifacts, training metadata, performance history	~500GB
Search / Intelligence	OpenSearch	2.x	Fraud indicator search, pattern library full-text search, investigation search	~300GB

17.2 Data Flow Architecture

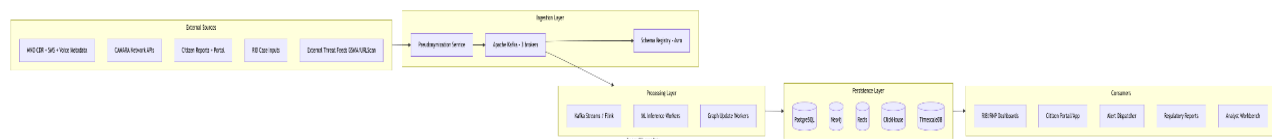


Figure 23: iNgabo Platform Data Flow Architecture

17.3 Data Classification and Handling

Table 37: Data Classification and Handling Requirements

Data Class	Examples	Encryption	Retention	Access Control
PUBLIC	Scam campaign alerts, fraud awareness content, public IMEI check results	TLS in transit only	Indefinite	No restriction
INTERNAL	Aggregated fraud statistics, anonymized analytics, platform configuration	AES-256 at rest + TLS in transit	5 years	Platform operators
CONFIDENTIAL	Fraud risk scores, behavioral profiles (pseudonymized), threat indicators	AES-256 + field-level encryption for quasi-identifiers	7 years (legal minimum)	Authorized services + analysts
RESTRICTED — PERSONAL	Citizen account data, MSISDN-linked records (hashed), case data	AES-256 + field-level encryption + HSM-managed keys	Per retention schedule + right to erasure	Named role + purpose + audit required
RESTRICTED — EVIDENCE	Chain of custody records, investigation data, CAMARA API responses	AES-256 + HSM-signed + write-once	10 years (court evidence standard)	RIB investigators + legal team + audit only
RESTRICTED — SECRETS	Encryption keys, API credentials, service certificates	HSM-only — never exported to disk	Per rotation schedule	Vault service only — no direct human access

17.4 Data Retention and Erasure

Table 38: Data Retention and Erasure Schedule

Data Category	Retention Period	Erasure Mechanism	Legal Basis
Fraud event records (anonymized)	7 years	Automated deletion after retention expiry	RURA regulatory requirement
Personal citizen data	5 years post-account closure OR right to erasure request	Cross-system deletion workflow triggered by citizen request or account closure	Rwanda Law 058/2021
Investigation / evidence data	10 years minimum (court evidence standard)	Court order or prosecution service authorization required for early deletion	Criminal Procedure Code
Audit log (immutable)	10 years	No deletion permitted — technical enforcement via ClickHouse write-once table	RURA + NCSA requirement
ML training data (anonymized)	Duration of model use + 2 years	Anonymization verification before archive; deletion on model retirement	Data minimization principle
CAMARA API response records	7 years (court evidence standard)	Evidence package closure + court authorization	RIB / legal framework
Behavioral profiles (pseudonymized)	7 years	Automated expiry + account erasure cascade	Law 058/2021 + RURA

18. GRAPH DATABASE DESIGN

SECTION 18

GRAPH DATABASE DESIGN

Neo4j Schema, Cypher Queries, and Fraud Network Intelligence



18.1 Design Rationale

Fraud is fundamentally a network phenomenon. Individual transactions, calls, or device events appear benign in isolation; their fraud nature emerges only in the context of relationships - who is calling whom, which device has been used with how many SIMs, which accounts are connected through a chain of transfers. Neo4j's native graph storage and query engine is architecturally the correct tool for this problem class; relational join-based approaches for the same queries are typically 100–1000× slower at the entity counts relevant to national-scale deployment.

18.2 Node Labels and Properties

Table 39: Neo4j Node Schema

Node Label	Key Properties	Index Strategy	Estimated Count (Phase 4)
:Citizen	citizen_id (UUID), risk_score, created_at, status	Unique index: citizen_id; range index: risk_score	~500K active citizens
:MSISDN	msisdn_hash (HMAC-SHA256), risk_score, fraud_flags[], last_seen, status	Unique index: msisdn_hash; composite: status + risk_score	~5M (all Rwanda MSISDNs)
:IMEI	imei_hash, status (ACTIVE/BLACKLISTED/UNDER_INVESTIGATION), device_model, first_seen	Unique index: imei_hash; index: status	~3M registered devices
:SIMCard	iccid_hash, mno, registration_date, status, risk_score	Unique index: iccid_hash; index: mno	~6M SIM cards (with history)
:Device	device_id, make, model, os, os_version	Unique index: device_id	~3M devices

Node Label	Key Properties	Index Strategy	Estimated Count (Phase 4)
:Location	cell_id, district, sector, lat_approx, lon_approx, mno	Unique index: cell_id + mno composite	~10K cell towers Rwanda
:FraudCase	case_id, status, severity, type, created_at, resolved_at	Unique index: case_id; index: status + severity	~50K cases (cumulative)
:Suspect	suspect_id, risk_level, aliases[], known_affiliations[]	Unique index: suspect_id; full-text: aliases	~5K identified suspects
:FraudCampaign	campaign_id, type, start_date, status, victim_count, estimated_loss_rwf	Unique index: campaign_id; index: status + type	~2K campaigns
:Account	account_hash, institution_code, account_type, risk_score, currency	Unique index: account_hash + institution_code composite	~2M accounts
:ThreatIndicator	ioc_id, type, value_hash, confidence, source, expires_at	Unique index: ioc_id; index: type + confidence	~1M active IOCs

18.3 Relationship Types and Properties

Table 40: Neo4j Relationship Schema

Relationship	From → To	Key Properties	Fraud Significance
OWNS	Citizen → MSISDN, Citizen → Account	since, status (ACTIVE/TRANSFERRED)	Ownership chain for account takeover analysis
USES_SIM	MSISDN → SIMCard	since, until, duration_days	SIM rotation frequency — high rotation = SIM farm indicator
INSERTED_IN	SIMCard → Device	first_seen, last_seen	Device-SIM pairing — multiple SIMs per device = fraud risk
HAS_IMEI	Device → IMEI	assigned_at	Device identity anchor
CALLED	MSISDN → MSISDN	timestamp, duration_sec, call_type, cell_id_origin	Communication graph — used for campaign detection and victimology
TRANSACTIONED_WITH	Account → Account	timestamp, amount_rwf, currency, channel, fraud_flag	Transaction graph — fan-in/fan-out pattern detection
LAST_SEEN_AT	MSISDN → Location, Device → Location	timestamp, signal_strength	Geographic intelligence and velocity anomaly detection
CONTROLS	Suspect → MSISDN, Suspect → Account	confidence, method (DIRECT/INFERRED)	Criminal attribution — basis for prosecution evidence

Relationship	From → To	Key Properties	Fraud Significance
PART_OF	Suspect → FraudCampaign	role (LEADER/OPERATOR/MULE), joined_at	Criminal organization structure
INVOLVES	FraudCase → MSISDN, FraudCase → IMEI, FraudCase → Account	role, added_at, added_by	Case scope management
LINKED_TO	FraudCase → FraudCase	link_type (SAME_CAMPAIGN/SAME_SUSPECT/SHARED_MODUS), confidence	Cross-case intelligence correlation
VICTIMIZED	FraudCampaign → Citizen	first_contact, last_contact, loss_rwf	Victim impact tracking

18.4 Representative Cypher Queries

18.4.1 Fraud Ring Detection

```
// Query: Find all MSISDNs reachable from a flagged number within 2 hops
// Used for: Investigation scope expansion when a fraud MSISDN is identified
MATCH (start:MSISDN {msisdn_hash: $target_hash})
CALL gds.bfs.stream(
  "fraud-graph",
  {sourceNode: id(start), maxDepth: 2,
   relationshipTypes: ["CALLED", "TRANSACTIONED_WITH", "CONTROLS"]}
)
YIELD nodeId, path
WITH gds.util.asNode(nodeId) AS connected_node
WHERE connected_node:MSISDN AND connected_node.risk_score > 400
RETURN connected_node.msisdn_hash AS msisdn,
       connected_node.risk_score AS risk_score,
       connected_node.fraud_flags AS flags
ORDER BY risk_score DESC
LIMIT 50

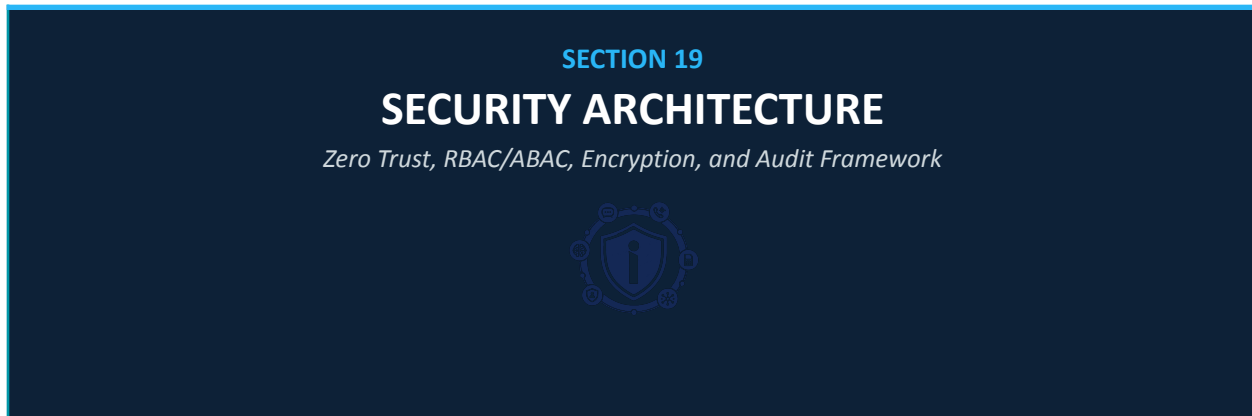
// Query: Detect money mule fan-in pattern
// Used for: Structured fraud / money laundering detection
MATCH (target:Account)
WHERE (datetime() - duration({hours: 24})) < datetime(target.last_active)
WITH target,
     size([(src:Account)-[:TRANSACTIONED_WITH]->(target)
          WHERE src.institution_code <> target.institution_code | src]) AS
unique_senders
WHERE unique_senders > 10
RETURN target.account_hash,
       target.risk_score,
       unique_senders
```

```
ORDER BY unique_senders DESC

// Query: Community detection – identify largest fraud cluster
// Used for: Campaign scope assessment and LEA briefing
CALL gds.louvain.stream("fraud-graph", {
  relationshipTypes: ["CALLED", "TRANSACTIONED_WITH", "CONTROLS", "PART_OF"],
  maxLevels: 10,
  gamma: 1.0
})
YIELD nodeId, communityId
WITH communityId, count(nodeId) AS community_size
ORDER BY community_size DESC
LIMIT 10
RETURN communityId, community_size
```

Figure 24: Representative Cypher Queries - Fraud Intelligence

19. SECURITY ARCHITECTURE



19.1 Zero Trust Architecture Implementation

iNgabo implements NIST SP 800-207 Zero Trust Architecture principles across all components. The core tenets — verify explicitly, use least privilege access, assume breach — are enforced architecturally rather than through policy alone. No implementation shortcut is accepted that undermines zero trust principles.

Table 41: Zero Trust Architecture Pillar Implementation

ZTA Pillar	NIST Requirement	iNgabo Implementation	Enforcement Point
Identity	Verify all users and services with strong authentication	Keycloak OIDC for humans (MFA mandatory); SPIFFE/SPIRE workload identity for services; certificate-based for MNO integrations	Keycloak + Istio mTLS policy
Device	Validate device health and compliance before granting access	MDM compliance check for admin and officer devices; PAW requirement for production admin; certificate validation for API clients	cert-manager + OPA policy
Network	Assume network is hostile; encrypt all traffic; enforce micro-segmentation	Istio service mesh with STRICT mTLS cluster-wide; Kubernetes NetworkPolicy; no plaintext permitted on any interface	Istio + NetworkPolicy
Application	Apply access controls at application layer; validate every request	OPA Gatekeeper for K8s admission; per-endpoint ABAC policy; signed JWT for every API request	Kong + OPA + Keycloak
Data	Protect data at rest and in transit; minimize data access by purpose	AES-256-GCM at rest (Vault-managed keys); TLS 1.3 in transit; field-level encryption for PII; pseudonymization at ingestion	Vault + PostgreSQL encryption + app layer

ZTA Pillar	NIST Requirement	iNgabo Implementation	Enforcement Point
Visibility	Monitor all activity; maintain complete audit trail	OpenTelemetry distributed tracing; Prometheus metrics; Loki structured logs; Splunk SIEM; immutable ClickHouse audit log	Istio telemetry + audit-log-service

19.2 Role-Based and Attribute-Based Access Control

19.2.1 Role Hierarchy

Table 42: RBAC Role Hierarchy

Role	Description	Access Scope	MFA Required
CITIZEN_ANON	Unauthenticated citizen	Public endpoints only: number check, scam report	No
CITIZEN_VERIFIED	MSISDN-verified citizen	Own case data; own SIM swap history; alerts management	SMS OTP
CITIZEN_FULL	Irembo-linked full account	All citizen functions; enhanced monitoring subscription	Irembo MFA
ANALYST_JUNIOR	Junior fraud analyst	Read-only access to fraud dashboard; own case queue; no personal data	TOTP app
ANALYST_SENIOR	Senior fraud analyst	Full analyst workbench; HITL review queue; case creation; pattern library management	TOTP app + hardware key
INVESTIGATOR_RIB	RIB cybercrime investigator	Full case management; evidence export; CAMARA API (with case ref); AI recommendations	Hardware key mandatory
INVESTIGATOR_RNP	RNP police officer	Dispatch module; case status; assigned case details; recovery outcome recording	Hardware key mandatory
DISPATCH_COORDINATOR	RNP dispatch coordinator	Dispatch module (full); unit management; case assignment; map intelligence	TOTP app
SUPERVISOR	Investigation supervisor	All investigator functions; case escalation; HITL sign-off for imminent threats; report approval	Hardware key mandatory
REGULATOR_RURA	RURA authorized staff	Regulatory compliance dashboard; fraud statistics; IMEI registry status	TOTP app
REGULATOR_BNR	BNR authorized staff	Financial fraud statistics; STR access; MoMo fraud dashboard	TOTP app

Role	Description	Access Scope	MFA Required
PLATFORM_ADMIN	iNgabo platform administrator	Platform configuration; user management; key rotation; audit access	Hardware key + PAW mandatory
SECURITY_AUDITOR	Third-party / internal auditor	Read-only audit log access; security metrics; no operational data	Hardware key + separate VPN

19.2.2 Attribute-Based Policies (ABAC)

RBAC alone cannot express all access control requirements. OPA (Open Policy Agent) implements ABAC policies that combine role, context, and data attributes for fine-grained control:

```
# OPA Policy Example: CAMARA Device Location API access control
package ingabo.camara.device_location

default allow = false

# Allow access only if ALL conditions are met:
allow {
  # 1. Caller has INVESTIGATOR_RIB role
  input.user.roles[_] == "INVESTIGATOR_RIB"

  # 2. A valid active case reference is provided
  input.case_ref != ""
  case := data.cases[input.case_ref]
  case.status == "ACTIVE"

  # 3. The requesting investigator is assigned to this case
  case.assigned_investigator == input.user.officer_id

  # 4. Case was authorized for location intelligence by supervisor
  case.location_intel_authorized == true

  # 5. Request is within business hours OR has emergency override
  within_authorized_hours
}

# Emergency override requires supervisor co-authorization
allow {
  input.user.roles[_] == "INVESTIGATOR_RIB"
  input.emergency_override.authorized_by_supervisor == true
  input.emergency_override.supervisor_id != input.user.officer_id
  time.now_ns() - input.emergency_override.authorized_at_ns < 3600000000000 # 1
  hour
}
```

Figure 25: OPA ABAC Policy — CAMARA Device Location Access Control

19.3 Encryption Architecture

Table 43: Encryption Architecture Summary

Encryption Context	Algorithm	Key Management	Key Rotation
Data at rest — PostgreSQL	AES-256-GCM (TDE via pgcrypto)	Vault Transit Engine → HSM root key	Annual + on compromise
Data at rest — Neo4j	AES-256 native encryption	Vault-managed data encryption key	Annual + on compromise
Data at rest — Kafka	AES-256 (broker-side encryption)	Vault-managed per-topic keys	Quarterly
Data at rest — ClickHouse	AES-256 encrypted columns for sensitive fields	Vault Transit Engine	Annual
Data in transit — all services	TLS 1.3 (TLS 1.2 minimum — older not permitted)	cert-manager + Let's Encrypt (external) / internal CA (internal)	90-day cert rotation (automated)
Service-to-service (mesh)	mTLS via Istio (SPIFFE x509 SVID)	Istio CA / SPIRE	24-hour cert rotation (automated)
Field-level encryption (PII)	AES-256-GCM application-layer encryption	Vault Transit Encryption-as-a-Service (EaaS)	Annual or on data subject request
HMAC pseudonymization (MSISDN/IMEI)	HMAC-SHA256 with per-deployment secret	Vault KV v2 — per-domain secrets	Annual rotation with re-pseudonymization
Signing (evidence packages)	RSA-4096 or ECDSA P-384	HSM-backed signing key — never exported	Annual with HSM ceremony
JWT signing (API auth tokens)	RS256 (RSA-SHA256)	Vault PKI — Keycloak signing keys	7-day rotation (automated)

19.4 Audit Log Architecture

All significant platform actions generate immutable audit events written to a dedicated ClickHouse audit table configured with write-once semantics (no UPDATE or DELETE permitted at the database level). Audit events are also published to a Kafka topic with separate consumer for out-of-band backup to an air-gapped audit archive.

```
// Audit Event Schema (Avro)
{
  "event_id": "AUD-2026-0001234567",    // UUID — globally unique
  "timestamp": "2026-06-15T14:23:11.422Z",
  "event_type": "CAMARA_DEVICE_LOCATION_QUERY",
```

```
"severity": "HIGH",
"actor": {
  "type": "HUMAN", // HUMAN | SERVICE | AUTOMATED
  "identity": "[officer_id_hash]",
  "role": "INVESTIGATOR_RIB",
  "session_id": "[session_uuid]",
  "ip_hash": "[IP pseudonymized]",
  "device_fingerprint": "[device cert hash]"
},
"resource": {
  "type": "CAMARA_API",
  "identifier": "[subscriber_ref_hash]",
  "camara_api": "DEVICE_LOCATION",
  "case_ref": "RIB-CYB-2026-0045"
},
"action": { "name": "QUERY", "outcome": "SUCCESS",
  "authorization": { "purpose_code": "LEA_INVESTIGATION",
    "policy_evaluated": "ingabo.camara.device_location", "decision": "ALLOW",
    "supervisor_case_ref": "SUP-AUTH-2026-089" }
},
"integrity": {
  "record_hash": "sha256:[hash of all above fields]",
  "chain_hash": "sha256:[hash of this record + previous record hash]"
}
}
```

Figure 26: Audit Event Schema — Tamper-Evident Chain Hash Design

20. PRIVACY AND COMPLIANCE ARCHITECTURE

SECTION 20

PRIVACY AND COMPLIANCE

Rwanda Law 058/2021, RURA, BNR, and International Standards



20.1 Privacy by Design Implementation

Privacy is architecturally enforced throughout iNgabo — it is not a policy layer added after system design. The following mechanisms implement Privacy by Design principles at the infrastructure and application level:

Table 44: Privacy by Design Architectural Implementations

Privacy Principle	Architectural Implementation
Data minimization	Ingestion contracts define maximum permissible fields per data type; additional fields rejected at API gateway; schema validation rejects over-provisioned payloads
Purpose limitation	Purpose codes enforce data use restrictions at API gateway (OPA) and application layer; data cannot be used for undeclared purposes without policy update and governance review
Pseudonymization by default	MSISDNs and IMEIs are hashed (HMAC-SHA256) at the ingestion boundary; plain-text identifiers never enter the core processing layer; re-identification requires authorized key retrieval logged to audit
Consent management	Citizen portal tracks explicit consent for each data processing activity; consent withdrawal triggers automated data restriction cascade; USSD/SMS interactions operate on implied consent (service delivery) only
Right to erasure automation	Citizen erasure request triggers cross-system deletion workflow spanning PostgreSQL, Redis, Neo4j (anonymization of nodes), ClickHouse (pseudonymization of retained audit records); completion report generated
Access minimization	Services receive only data necessary for their function — enforced by field-level API response filtering, not trust; the fraud-detection-engine cannot query citizen personal details
Data separation	Personal identity data and fraud intelligence data are stored in separate database schemas with no direct joins; correlation requires explicit service call through auth-controlled API

20.2 Compliance Mapping

Table 45: Regulatory Compliance Mapping

Requirement	Source Regulation	iNgabo Implementation	Evidence Artifact
Lawful basis for processing	Law 058/2021 Art. 7	Data Sharing Agreements (RURA-authorized); citizen consent for portal; legitimate interest analysis for fraud prevention	Legal register + DPA review
Data Protection Officer	Law 058/2021 Art. 35	Platform DPO designated; DPO has access to all data processing records and can suspend data flows	DPO appointment letter
DPIA for high-risk processing	Law 058/2021 Art. 32	DPIA completed for: CDR processing, CAMARA API usage, behavioral profiling, voice metadata analysis	DPIA documentation
Data subject rights	Law 058/2021 Arts. 17–26	Automated rights fulfillment workflow (access, erasure, rectification, portability) with 30-day response SLA	Rights management system
Data breach notification	Law 058/2021 Art. 39	72-hour breach notification to DPA; subscriber notification within 7 days if at-risk; automated breach detection via SIEM	Incident response runbook
Cross-border transfer	Law 058/2021 Art. 41	Standard contractual clauses for EAC integration; adequacy assessment per destination country (Phase 5)	Transfer impact assessments
Telecom data sharing	RURA Electronic Communications Act	Formal Data Sharing Agreements with each MNO; RURA audit compliance reporting	DSA register + RURA monthly report
Financial data processing	BNR Mobile Money Policy	BNR authorized processing agreement; STR generation in BNR-specified format; FIU access controls	BNR agreement + STR audit log
Cybersecurity standards	NCSA National Cybersecurity Policy	NCSA penetration test clearance; security incident notification within 4 hours; threat intelligence sharing	NCSA assessment report
AI accountability	Emerging Rwanda AI Policy 2025	XAI mandatory on all risk scores; HITL for enforcement; bias audit quarterly; algorithm register maintained	AI governance report

21. DEPLOYMENT ARCHITECTURE

SECTION 21

DEPLOYMENT ARCHITECTURE

Kubernetes, GitOps, CI/CD, and Infrastructure as Code



21.1 Infrastructure as Code

All iNgabo infrastructure is defined as code using Terraform for cloud/infrastructure provisioning and Helm charts for Kubernetes application deployment. No manual infrastructure changes are permitted in production environments; all changes flow through the GitOps pipeline with mandatory code review, automated security scanning, and audit logging.

Table 46: Infrastructure as Code Toolchain

IaC Component	Tool	Repository	Deployment Method
Cloud / bare-metal infrastructure	Terraform	ingabo-infra-tf	Terraform Cloud / atlantis with PR-based apply
Kubernetes cluster configuration	Terraform + kubeadm	ingabo-infra-tf	Automated cluster bootstrap with kubeadm
Application deployments	Helm 3 + ArgoCD	ingabo-helm-charts	GitOps: ArgoCD watches chart repo, auto-syncs
Service mesh (Istio) configuration	Istio Operator + Helm	ingabo-helm-charts/istio	ArgoCD managed — Istio resource CRDs
Secret seeding (initial)	Vault CLI + Terraform	ingabo-vault-config	Manual ceremony for initial HSM setup; automated after
Network policies	Kubernetes manifests (YAML)	ingabo-helm-charts	ArgoCD managed
Monitoring stack	Helm (kube-prometheus-stack, Loki)	ingabo-helm-charts/monitoring	ArgoCD managed
Security policies (OPA)	Rego policies + Helm	ingabo-opa-policies	ArgoCD managed — policy evaluation auto-tested in CI

21.2 CI/CD Pipeline

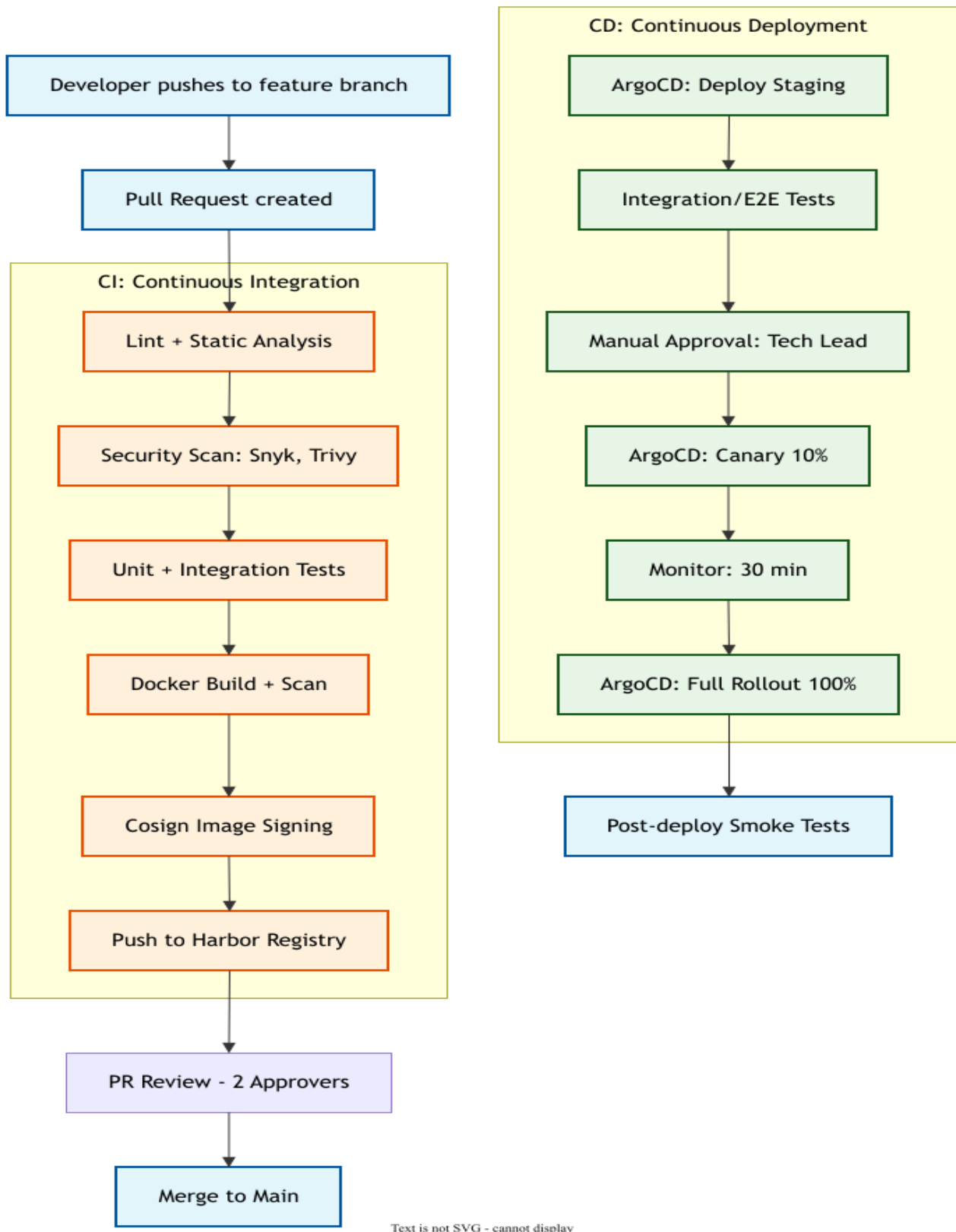


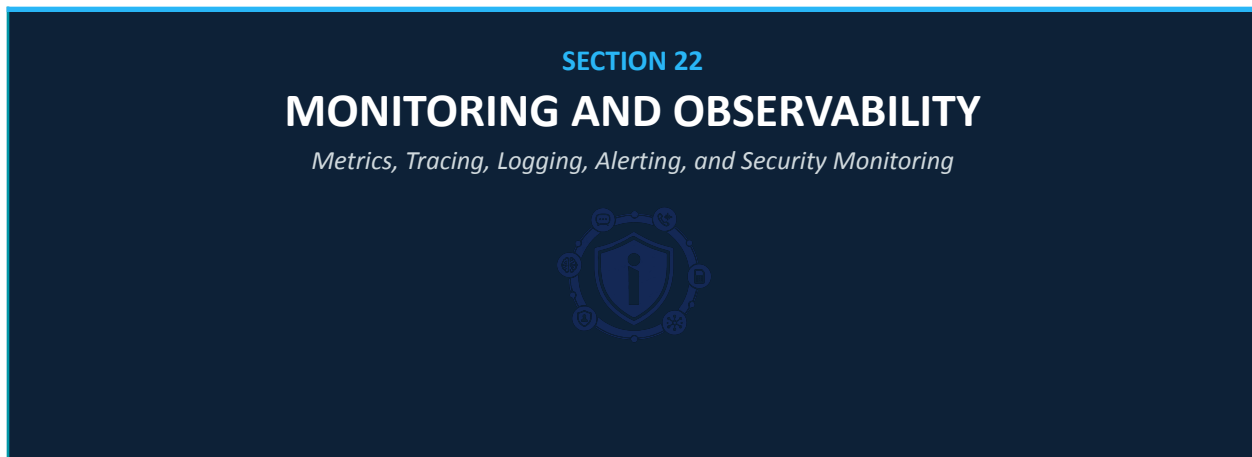
Figure 27: iNgabo CI/CD Pipeline - GitOps with Security Gates

21.3 Environment Strategy

Table 47: Deployment Environment Strategy

Environment	Purpose	Data	Access Control	Refresh Strategy
Development (dev)	Active feature development	Fully synthetic data — no real subscriber data	Developer team only	On-demand per feature branch
Staging	Integration testing, UAT, pre-production validation	Anonymized synthetic data resembling production patterns	Dev + QA + security team	Weekly full refresh; on-demand for releases
Production	Live national platform	Real operational data — full security controls applied	Operators and authorized users per RBAC	GitOps canary deployment
DR / Standby	Disaster recovery failover	Real-time replication from production	Same as production — separate credentials	Continuous replication; test monthly
Security Sandbox	Penetration testing, security research	Isolated clone of staging — no production connectivity	Security team + authorized third-party auditors	On-demand per security engagement

22. MONITORING AND OBSERVABILITY



22.1 Observability Architecture

iNgabo implements the three pillars of observability — metrics, tracing, and logs — using the OpenTelemetry standard for instrumentation, enabling vendor-neutral observability data collection across all services.

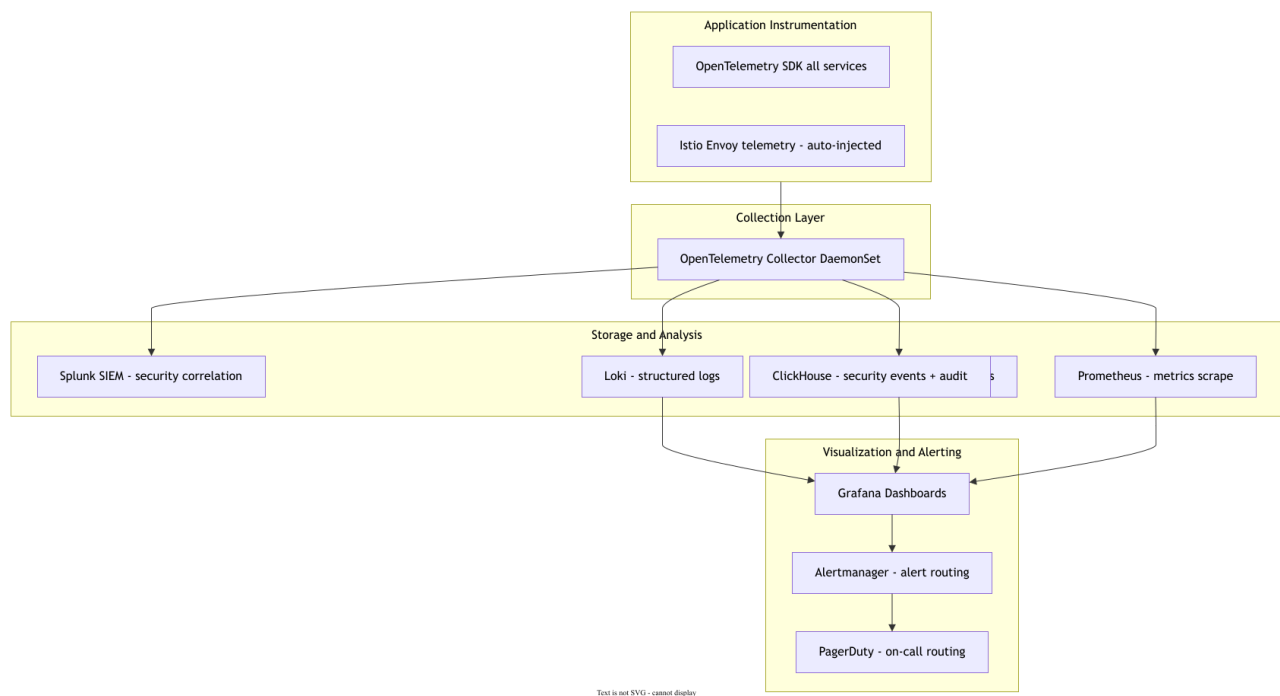


Figure 28: iNgabo Observability Stack Architecture

22.2 Key Performance Indicators — Dashboard Design

Table 48: Operational Dashboard Design

Dashboard	Audience	Key Metrics	Refresh Rate
Fraud Detection Operations	Fraud analysts, platform operators	Events/sec ingested; fraud alerts/hr; false positive rate; HITL queue depth; alert delivery latency; campaign detections active	30 seconds
RIB Investigation Operations	RIB operations leads	Open cases by status/priority; HITL actions pending; average case resolution time; evidence export count; dispatch response time	1 minute
Platform Health	Platform SRE team	Service availability per microservice; API response time P50/P95/P99; Kafka consumer lag; database connection pool; K8s pod health	15 seconds
Security Operations	Security team + NCSA	Auth failures; anomalous API access patterns; OPA policy violations; mTLS handshake failures; Vault unsealing events; audit log integrity alerts	Real-time
AI Model Performance	ML engineering team	Model inference latency; prediction confidence distribution; feature drift metrics; HITL override rate (human vs AI agreement); false positive trend	1 hour (batch KPI update)
Regulatory Compliance	RURA, BNR (authorized access)	Monthly fraud statistics; IMEI blacklist count; SIM swap intercepts; STR submissions; data sharing agreement API call counts	Daily batch refresh
Executive Summary	MINICT, RIB leadership	Total fraud prevented (estimated RWF); citizen portal users; alert volume; recovery rate; platform uptime; regional expansion status	Daily

22.3 Alerting Strategy

Table 49: Alerting Strategy and Escalation Matrix

Alert Category	Condition	Severity	On-Call Response	Escalation
Fraud Detection Failure	Kafka consumer lag > 10,000 events on fraud-events topic for > 2 minutes	P1 CRITICAL	Immediate page — SRE on-call	Automatic: escalate to SRE lead + Platform CTO at 5 min
SIM Swap Gateway Down	simswap-gateway service unavailable > 30 seconds	P1 CRITICAL	Immediate page — SRE on-call	Automatic: MNO partners notified; RIB + RURA notification at 2 min

Alert Category	Condition	Severity	On-Call Response	Escalation
API Gateway Error Rate High	Kong 5xx error rate > 1% over 5 minutes	P2 HIGH	Page SRE within 5 min	Escalate to SRE lead at 15 min
ML Inference Latency	ml-inference-service P99 > 800ms over 10 minutes	P2 HIGH	Page ML ops within 15 min	Escalate to ML engineering lead at 30 min
Fraud Alert Delivery Failure	alert-dispatcher delivery failure rate > 5% over 10 minutes	P2 HIGH	Page SRE + operations within 15 min	RURA notification if > 30 min
Database Replication Lag	PostgreSQL replica lag > 60 seconds	P3 MEDIUM	Notify DBA on-call within 30 min	Escalate if lag increases over 15 min
Certificate Expiry Warning	Any TLS certificate expiring within 14 days	P3 MEDIUM	Notify SRE team via email	Escalate if cert not renewed within 7 days
Unauthorized API Access Attempt	OPA policy denial rate > 10/min from single source	P2 HIGH	Security team page within 10 min	Automatic IP block after 50 denials; NCSA notification
Audit Log Gap	Gap detected in ClickHouse audit chain hash sequence	P1 CRITICAL	Immediate security team + CISO page	NCSA notification within 1 hour; legal team briefed
HSM Unavailable	Vault HSM connection failure	P1 CRITICAL	Immediate SRE + Security page	Platform goes to read-only mode; CISO + legal briefed immediately

23. SCALABILITY STRATEGY

SECTION 23

SCALABILITY STRATEGY

Horizontal Scaling, Auto-Scaling, and National-to-Continental Growth



23.1 Horizontal Scaling Architecture

Every iNgabo service is designed for horizontal scaling. Stateful services (databases) scale through clustering and sharding; stateless services (APIs, ML inference, stream processors) scale through Kubernetes Horizontal Pod Autoscaler (HPA) based on CPU, memory, and custom metrics (Kafka consumer lag, inference queue depth).

Table 50: Component Horizontal Scaling Configuration

Component	Scaling Mechanism	Trigger Metric	Scale-Out Time	Max Scale Target
fraud-detection-engine	K8s HPA + Kafka consumer group scaling	Kafka: fraud-events consumer lag > 5,000	< 90 seconds (new pod + warm-up)	50 pods (national scale)
ml-inference-service	K8s HPA (GPU nodes)	Custom metric: inference queue depth > 100	< 3 minutes (GPU node provisioning)	20 GPU pods
nlp-classifier	K8s HPA	CPU > 70% OR queue depth > 500	< 90 seconds	30 pods
graph-intelligence-engine	K8s HPA + Neo4j read replica	Custom metric: graph query queue > 50	< 2 minutes	15 pods + 5 Neo4j read replicas
simswap-gateway	K8s HPA	RPS > 500 per pod	< 60 seconds	20 pods
citizen-api	K8s HPA	CPU > 60% OR RPS > 200 per pod	< 60 seconds	30 pods
alert-dispatcher	K8s HPA + Kafka consumer scaling	Kafka: fraud-alerts consumer lag > 1,000	< 90 seconds	20 pods
Kafka brokers	Manual scaling (stateful — planned) + partition rebalancing	Storage > 70% OR network saturation	Manual — planned quarterly review	10 brokers (continental scale)

Component	Scaling Mechanism	Trigger Metric	Scale-Out Time	Max Scale Target
Neo4j	Add read replicas (causal cluster)	Query latency P95 > 2 seconds	~5 minutes (follower catch-up)	1 leader + 10 followers
PostgreSQL	Read replicas via PgBouncer	Read query CPU > 70% on primary	~3 minutes	1 primary + 5 replicas per shard

23.2 Multi-Tenant Regional Architecture (Phase 5+)

From Phase 5, iNgabo operates as a multi-tenant federated platform where each EAC member state deployment maintains its own data sovereignty while contributing to and consuming from a shared fraud intelligence graph. The multi-tenant architecture isolates data at the namespace and database schema level:

Table 51: Multi-Tenant Data Isolation Architecture

Tenant Isolation Layer	Mechanism	Data Sovereignty Guarantee
Kubernetes namespace	Each member state = dedicated K8s namespace with NetworkPolicy isolation	No cross-namespace data flow without explicit federation API call
Database schema	Each member state = dedicated PostgreSQL schema in shared cluster OR dedicated cluster (high-sovereignty mode)	Schema-level row security prevents cross-tenant data access
Neo4j tenant property	All nodes tagged with tenant_id property; query-time row-level security filter	Graph queries cannot traverse cross-tenant relationships without federation intent
Kafka topic namespace	All topics prefixed with tenant code (e.g., RW_, UG_, KE_)	Cross-topic access requires explicit federation consumer group with audit
Encryption key isolation	Each tenant has dedicated Vault namespace with independent key hierarchy	Cross-tenant re-identification technically impossible without both key sets
Intelligence federation API	Standardized STIX 2.1 feed published from each tenant; consumed by federation hub	Raw subscriber data never leaves tenant boundary — intelligence signals only

24. DISASTER RECOVERY ARCHITECTURE

SECTION 24

DISASTER RECOVERY

RTO, RPO, Backup Strategy, and Failover Procedures



24.1 Recovery Objectives

Table 52: Recovery Time and Point Objectives by Component

Component Class	RTO	RPO	Recovery Method
Fraud Detection Core (FDE, SSG)	< 5 minutes	Zero (event replay from Kafka)	Active-active multi-AZ; automatic K8s pod rescheduling
SIM Swap Gateway	< 1 minute	Zero	Active-active; Istio circuit breaker; load balancer health check
Citizen Portal	< 15 minutes	< 5 minutes	Multi-AZ + CDN fallback; static content served from CDN during backend recovery
RIB Investigation Dashboard	< 30 minutes	< 5 minutes	Multi-AZ replica; read-only mode on DR replica during primary recovery
PostgreSQL (operational data)	< 15 minutes	< 1 minute (synchronous replica)	Synchronous streaming replication; auto-promote replica on primary failure
Neo4j Graph Database	< 30 minutes	< 1 minute (Raft consensus)	Causal cluster; automatic leader election; follower promotion
Kafka Event Bus	< 10 minutes	Zero (durable log on disk)	3-broker cluster; automatic partition leadership rebalancing
ClickHouse Analytics	< 2 hours (analytics are not time-critical)	< 15 minutes	Async replication to DR replica; analytics acceptable to lag during recovery
HashiCorp Vault (HSM)	< 1 hour	Zero (sealed — no data loss possible)	Vault HA cluster; automatic unseal with quorum; HSM backup at secondary site

24.2 Backup Strategy

Table 53: Backup Strategy Summary

Data Store	Backup Method	Frequency	Retention	Restore Tested
PostgreSQL	pg_basebackup + WAL streaming to DR site	Continuous (WAL) + weekly full	90 days full; 7 years WAL for compliance	Monthly full restore test to staging
Neo4j	Online backup to S3-compatible (MinIO) + daily full dump	Daily full + hourly incremental	90 days	Monthly restore test
Kafka	Topic data retained on disk (replication factor 3) + weekly snapshot to cold storage	Continuous (replication) + weekly snapshot	90 days hot; 7 years cold (ClickHouse archive)	Quarterly DR simulation
ClickHouse	Replication + S3-compatible backup	Daily incremental + weekly full	7 years (regulatory requirement)	Quarterly full restore verification
Vault secrets	Vault auto-snapshot to encrypted S3 + cross-site replication	Hourly snapshot	30 days rolling	Quarterly restore test with HSM ceremony
ML models (MinIO)	S3 cross-region replication	Continuous (object replication)	All model versions retained	On new model deployment — verify restore
K8s etcd (cluster state)	etcd snapshot to S3	Every 30 minutes	30 days	Monthly restore test

24.3 DR Drill Schedule

Table 54: Disaster Recovery Drill Schedule

Drill Type	Frequency	Scope	Success Criteria
Database failover drill	Monthly	PostgreSQL primary failure; replica promotion; application reconnect	< 15 min RTO; zero data loss confirmed
Full platform DR simulation	Quarterly	Simulate complete primary AZ failure; full failover to DR AZ	All P1 services recovered within RTO; RPO verified
Backup restore verification	Monthly (PostgreSQL); Quarterly (full stack)	Restore from backup to isolated test environment; data integrity check	100% data integrity; RTO within target
Vault DR recovery	Quarterly	Simulate Vault cluster loss; recover from snapshot with HSM ceremony	Vault recovered and unsealed within 1 hour
Business continuity tabletop	Semi-annually	Walk through major incident scenarios (ransomware, data center fire, etc.) with all leads	Documented runbook gaps identified and resolved

25. PERFORMANCE TARGETS AND CAPACITY PLANNING

SECTION 25

PERFORMANCE TARGETS

Latency, Throughput, and National-Scale Capacity



25.1 Latency Targets

Table 55: Platform Latency Targets

Operation	P50 Target	P95 Target	P99 Target	Measurement Point
Fraud risk score (API)	< 150ms	< 500ms	< 1.5s	Kong API gateway response
SIM swap risk assessment	< 300ms	< 800ms	< 1.5s	simswap-gateway response to MNO webhook
SMS classification	< 100ms	< 300ms	< 800ms	nlp-classifier response
Graph fraud query (2-hop)	< 500ms	< 2s	< 4s	graph-intelligence-engine response
Citizen number verification (web)	< 800ms	< 2s	< 4s	End-to-end: browser to response
USSD response	< 1s	< 2.5s	< 5s	USSD session response (Africa's Talking SLA)
Investigation dashboard load	< 2s	< 4s	< 8s	Full page including graph render
Evidence export (PDF)	< 10s	< 30s	< 60s	Export service response (file ready)
Real-time fraud alert (end-to-end)	< 500ms	< 2s	< 5s	Event ingestion to citizen SMS dispatch
ML model inference (per event)	< 50ms	< 200ms	< 500ms	ml-inference-service internal response

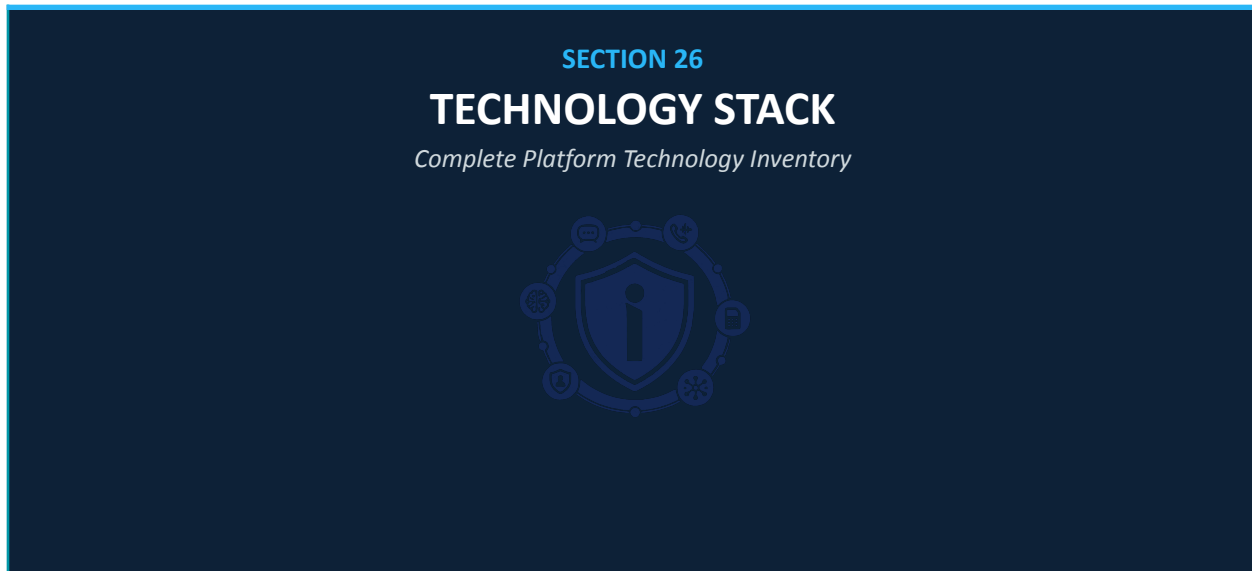
25.2 Throughput and Capacity

Table 56: Platform Capacity Targets by Phase

Capacity Dimension	Phase 2 Baseline	Phase 4 National	Phase 6 Continental
Fraud events processed/second	5,000	50,000	500,000

Capacity Dimension	Phase 2 Baseline	Phase 4 National	Phase 6 Continental
Risk score API requests/day	500K	10M	500M
SMS messages classified/day	2M	20M	500M
SIM swap checks/day	10K	100K	5M
Active Neo4j nodes (graph)	500K	10M	500M
Active Neo4j relationships	2M	100M	5B
Kafka message throughput	100K msg/sec	1M msg/sec	50M msg/sec
Concurrent citizen portal users	1,000	50,000	500,000
Concurrent investigator sessions	20	500	5,000
ClickHouse query QPS	100	2,000	20,000

26. TECHNOLOGY STACK



26.1 Backend Services

Table 57: Backend Technology Stack

Technology	Version	License	Use Case	Justification
NestJS (Node.js/TypeScript)	10.x	MIT	REST API services, USSD gateway, alert dispatcher, regulatory reporting	TypeScript-first; built-in DI, guards, interceptors; decorator-based middleware; strong enterprise patterns
Python / FastAPI	3.11 / 0.100+	MIT	AI/ML inference services, NLP engine, fraud detection engine	Leading ML ecosystem; async-first FastAPI; strong scientific Python community
Apache Kafka	3.5 (KRaft)	Apache 2.0	Event streaming bus — all inter-service async communication	Industry standard for event streaming; proven at billion-event scale; KRaft removes ZooKeeper dependency
Apache Spark	3.4	Apache 2.0	Batch feature engineering, nightly behavioral baseline computation	Industry standard for large-scale batch ML feature pipelines
gRPC	1.55+	Apache 2.0	Synchronous inter-service calls (ML inference, SIM swap)	Protobuf-based; strongly typed; bidirectional streaming; 5–10× faster than REST for internal calls
GraphQL (Apollo)	4.x	MIT	Flexible client query API for investigation dashboard	Reduces over-fetching for complex investigation data;

Technology	Version	License	Use Case	Justification
				enables efficient frontend queries

26.2 AI and Machine Learning Stack

Table 58: AI/ML Technology Stack

Technology	Version	License	Use Case
TensorFlow / Keras	2.13+	Apache 2.0	Deep neural network fraud classifier, LSTM temporal model, Autoencoder behavioral anomaly
XGBoost	1.7+	Apache 2.0	Primary fraud classification model (tabular CDR features) — production-proven in fraud domain
scikit-learn	1.3+	BSD	Feature engineering pipelines, Isolation Forest anomaly detection, preprocessing
PyTorch / DGL / PyTorch Geometric	2.0+ / 1.1+ / 2.3+	BSD / Apache 2.0	Graph Neural Network training and inference on Neo4j-exported subgraphs
HuggingFace Transformers	4.30+	Apache 2.0	mBERT fine-tuning for SMS fraud classification; KinyaBERT for Kinyarwanda NLP
TensorFlow Serving	2.13+	Apache 2.0	Production ML model serving with versioning and A/B traffic splitting
SHAP	0.42+	MIT	Explainable AI — SHapley value computation for all production model outputs
MLflow	2.5+	Apache 2.0	Experiment tracking, model registry, model versioning, deployment lifecycle
Evidently AI	0.4+	Apache 2.0	ML model drift monitoring in production — feature drift, prediction drift, data quality
Fairlearn	0.9+	MIT	AI bias detection and mitigation for fraud scoring models
Label Studio	1.7+	Apache 2.0	Fraud data annotation platform for training data quality
MinHash / datasketch	1.1+	MIT	LSH-based SMS campaign fingerprinting for near-duplicate detection

26.3 Database and Storage

Table 59: Database and Storage Technology Stack

Technology	Version	License	Use Case
PostgreSQL	15+	PostgreSQL License	Primary operational database — cases, citizens, configurations, operational state

Technology	Version	License	Use Case
TimescaleDB	2.11+ (PG extension)	Apache 2.0	Behavioral time-series data — per-entity baselines and anomaly metrics
Neo4j Enterprise	5.x	Commercial	Fraud intelligence graph — entities, relationships, GNN inference, Cypher queries
Redis	7.x (Cluster mode)	BSD	Real-time cache — risk scores, blacklists, session data, rate limiting, pub/sub
Apache Kafka	3.5	Apache 2.0	Event streaming bus — all platform events retained for 90 days
ClickHouse	23.x (Cluster)	Apache 2.0	Fraud event analytics, immutable audit log, behavioral analytics, regulatory reports
MinIO	RELEASE.2023	AGPL v3 (self-hosted)	S3-compatible object storage — ML models, backups, evidence packages, exports
OpenSearch	2.x	Apache 2.0	Fraud indicator full-text search, pattern library, investigation search
Schema Registry (Confluent)	7.4 (open source)	Community License	Avro schema management for Kafka topics

26.4 Frontend and Mobile

Table 60: Frontend and Mobile Technology Stack

Technology	Version	License	Use Case
React.js	18.x	MIT	Citizen web portal, RIB investigation dashboard, analyst workbench (PWA-ready)
React Native	0.72+	MIT	iOS and Android citizen mobile app - shared codebase for both platforms
D3.js / Sigma.js	7.x / 2.x	BSD / MIT	Fraud network graph visualization in investigation dashboard
Apache ECharts / Recharts	5.x / 2.x	Apache 2.0	Fraud analytics charts, heat maps, trend visualizations
TailwindCSS	3.x	MIT	Utility-first CSS for consistent design system across platforms
i18next	23.x	MIT	Internationalization: Kinyarwanda, English, French for all interfaces
MapLibre GL JS	3.x	BSD	Crime heat maps and dispatch location maps (open-source, no API key required)
PWA Workbox	7.x	MIT	Progressive Web App offline support and service worker management

26.5 Infrastructure and DevOps

Table 61: Infrastructure and DevOps Technology Stack

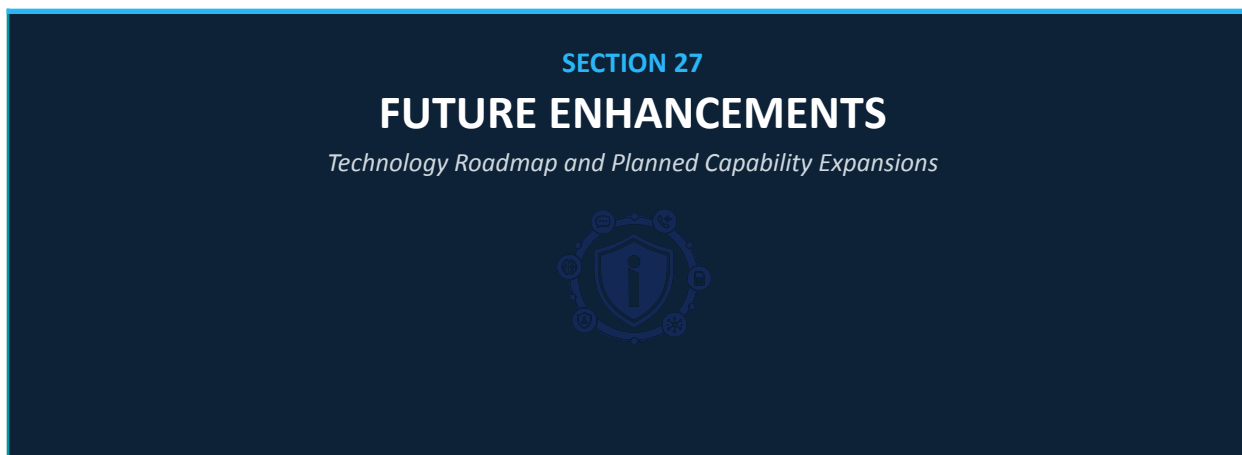
Technology	Version	License	Use Case
Kubernetes (K8s)	1.28+	Apache 2.0	Container orchestration — all production workloads
Docker	24.x	Apache 2.0	Container runtime — all services packaged as OCI-compliant containers
Istio	1.18+	Apache 2.0	Service mesh — mTLS, traffic management, telemetry, circuit breaking
Helm	3.12+	Apache 2.0	Kubernetes application packaging and deployment
ArgoCD	2.8+	Apache 2.0	GitOps continuous deployment — all production changes via ArgoCD
Terraform	1.5+	BUSL 1.1	Infrastructure as Code — cloud resources, networking, DNS, storage
GitLab CI/CD	Self-hosted	MIT (CE)	CI pipeline — build, test, security scan, image publish
Harbor	2.8+	Apache 2.0	Private container registry — image storage, vulnerability scanning, signing
Cosign / Sigstore	2.x	Apache 2.0	Container image signing and supply chain integrity verification
Kong	3.3+	Apache 2.0 (OSS)	API Gateway — rate limiting, authentication, WAF, routing, analytics
HashiCorp Vault	1.14+	BUSL 1.1	Secrets management — all platform secrets, encryption keys, PKI
cert-manager	1.12+	Apache 2.0	Automated TLS certificate management — Let's Encrypt + internal CA
Open Policy Agent (OPA)	0.54+	Apache 2.0	Policy as code — ABAC enforcement, K8s admission control
Falco	0.35+	Apache 2.0	Container runtime security monitoring — anomalous process detection
Keycloak	22+	Apache 2.0	Identity provider — OIDC/OAuth2, RBAC, MFA, federated identity
Africa's Talking	API v1	Commercial	USSD gateway, SMS delivery — Rwanda MNO coverage

26.6 Observability and Security

Table 62: Observability and Security Toolchain

Technology	Version	License	Use Case
Prometheus	2.45+	Apache 2.0	Metrics collection and storage — all service and infrastructure metrics
Grafana	10.x	AGPL v3	Metrics visualization, alerting, dashboard management
Grafana Loki	2.8+	AGPL v3	Centralized structured log aggregation
Grafana Tempo	2.2+	Apache 2.0	Distributed trace storage and query
OpenTelemetry Collector	0.82+	Apache 2.0	Unified telemetry collection — metrics, traces, logs from all services
Alertmanager	0.26+	Apache 2.0	Alert routing, grouping, deduplication, PagerDuty integration
Wazuh	4.5+	GPLv2	SIEM/XDR — host intrusion detection, log correlation, compliance
Snyk	Latest	Commercial (OSS free tier)	SAST, dependency scanning, container vulnerability scanning in CI/CD
Trivy	0.44+	Apache 2.0	Container and K8s manifest vulnerability scanning in CI/CD
OWASP ZAP	2.13+	Apache 2.0	Automated DAST — web API security testing in CI/CD
Falco	0.35+	Apache 2.0	Runtime container security — process anomaly, file access, network calls

27. FUTURE ENHANCEMENTS



27.1 Near-Term Enhancements (Phase 3–4, 2027–2028)

Table 63: Near-Term Enhancement Roadmap (Phase 3–4)

Enhancement	Description	Dependency	Expected Impact
Federated Learning for ML	Train fraud models across MNO data without centralizing raw data — privacy-preserving AI	MNO technical cooperation agreement; FL framework deployment (PySyft/FATE)	10–15% model accuracy improvement with 3× more training data; MNO data sharing without privacy compromise
SS7/DIAMETER signaling intelligence	Direct signaling protocol monitoring for telecom-layer fraud (IMSI catching, MAP abuse)	RURA authorization; MNO signaling access agreement; dedicated probes	Detection of 8 additional fraud attack classes invisible to CDR-only analysis
Real-time MoMo fraud prevention	Per-transaction risk scoring API integrated into MoMo transaction flow (pre-authorization)	BNR authorization; MoMo provider API integration; < 300ms latency SLA	Real-time prevention instead of post-event detection for mobile money fraud
Voice content analysis (authorized)	AI analysis of voice call content for vishing detection within RURA/court-authorized legal framework	Court authorization; RURA approval; technical MNO integration	Direct vishing script detection; social engineering call identification
Digital wallet integration	Integration with Rwanda's national digital identity wallet for highest-confidence KYC verification	Rwanda NDID wallet API availability; MINICT partnership	Near-zero false positives for identity verification; eliminate impersonation fraud
Biometric SIM swap challenge	Real-time biometric liveness challenge for high-risk SIM swap requests	MNO technical integration for biometric challenge in	Eliminates social engineering as SIM

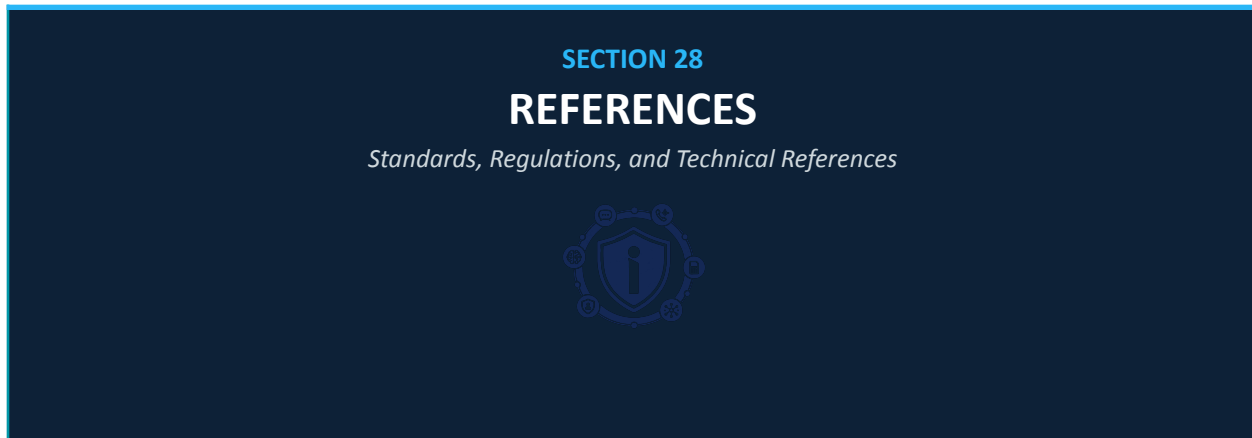
Enhancement	Description	Dependency	Expected Impact
		swap workflow; subscriber enrollment	swap vector for enrolled subscribers

27.2 Long-Term Vision (Phase 5–6, 2029–2031)

Table 64: Long-Term Vision Enhancement Roadmap (Phase 5–6)

Enhancement	Description	Continental Impact
5G Network Slicing Security	Dedicated 5G network slice for fraud detection with guaranteed latency and isolation	Sub-100ms fraud prevention response time; isolated secure channel for intelligence exchange
Quantum-Resistant Cryptography	Migrate all cryptographic protocols to post-quantum algorithms (CRYSTALS-Kyber, CRYSTALS-Dilithium)	Future-proof platform against quantum computing threats to public-key encryption
IoT Fraud Intelligence	Extend platform to detect identity fraud in IoT device provisioning (smart city, M2M)	Protects Rwanda's smart city infrastructure and IoT device economy from identity abuse
AI Fraud Prediction (Pre-Crime)	Predictive modeling to identify entities likely to be targeted before attack occurs	Shift from protective-reactive to fully predictive-protective model — highest maturity level
Blockchain Evidence Registry	Distributed ledger for cross-border court evidence admissibility across EAC jurisdictions	Eliminates legal barriers to cross-border prosecution of fraud ring operators
Pan-African Fraud Intelligence Exchange	Real-time STIX 2.1 federation across 15+ AU member state deployments	Continental fraud network disruption; first-ever pan-African organized crime digital intelligence capability

28. REFERENCES



28.1 Standards and Frameworks

Table 65: Standards and Frameworks References

Reference	Title	Relevance to iNgabo
IEEE 29148:2018	Systems and Software Engineering — Life Cycle Processes — Requirements Engineering	Requirements specification methodology for this TAD and companion SRS
ISO/IEC 25010:2023	Systems and Software Quality Requirements and Evaluation — Product Quality Model	Quality attribute framework for non-functional requirements specification
NIST SP 800-207	Zero Trust Architecture	ZTA design principles implemented throughout platform security architecture
NIST SP 800-53 Rev 5	Security and Privacy Controls for Information Systems	Control baseline for platform security architecture
NIST Cybersecurity Framework 2.0	Identify, Protect, Detect, Respond, Recover	Security lifecycle framework for platform security program
3GPP TS 33.117	Catalogue of General Security Assurance Requirements	Security requirements for telecom integration components
3GPP TS 33.250	Security Aspects of 5G Fraud Detection	5G fraud detection architecture reference for Phase 5
GSMA FS.07	SS7 Interconnect Security Monitoring and Firewall Guidelines	SS7 fraud detection methodology reference
GSMA FS.11	DIAMETER Interconnect Security Monitoring and Firewall Guidelines	DIAMETER fraud detection methodology reference
GSMA FS.38	GSMA Fraud Classification System	Fraud taxonomy alignment for pattern library classification

Reference	Title	Relevance to iNgabo
GSMA T-ISAC	Telecom-ISAC Threat Intelligence Standards	Intelligence sharing format and protocol reference
STIX 2.1 / TAXII 2.1	Structured Threat Information Expression / Transport	Threat intelligence interchange format for hub and federation
MITRE ATT&CK for Mobile	Mobile Threat Matrix	Attack taxonomy for mobile and telecom threat modeling
OWASP API Security Top 10	API Security Guidelines	API security design baseline
Linux Foundation CAMARA	Cloud and Network Automation Standardized APIs	CAMARA API integration specification and capability reference
ISO/IEC 27001:2022	Information Security Management Systems	ISMS framework for platform security governance
ISO/IEC 27035-1:2023	Information Security Incident Management	Incident response framework reference
NIST SP 800-188	De-Identification of Government Datasets	Pseudonymization methodology guidance

28.2 Rwanda Regulatory References

Table 66: Rwanda Regulatory References

Reference	Full Title	Relevance
Law No. 058/2021	Law on the Protection of Personal Data and Privacy in Rwanda	Primary data protection compliance framework
Law No. 24/2016	Law Governing Information and Communication Technologies	Telecom regulation framework
RURA Framework 2023	Telecom Fraud Reporting and SIM Registration Requirements (RURA Directive)	SIM swap regulation; IMEI registry; CDR sharing framework
BNR Directive No. 04/2020	Directive on Electronic Payment Instruments and Systems	Mobile money fraud detection requirements
Rwanda National Cybersecurity Policy 2024	National Cybersecurity Strategy and Policy Framework	NCSA requirements; incident reporting; intelligence sharing
Rwanda AI Policy 2025 (Draft)	National AI Governance Policy and Framework	AI governance requirements; XAI; algorithmic accountability

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